Higher-dimensional exponential metrics
A proposed axiomatic existence and uniqueness proof

Research manuscript

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Abstract

This manuscript presents a proposed proof of axiomatic existence and uniqueness for subcritical exponential metrics of the whole-space log-correlated Gaussian field in every dimension $d \geq 3$. Its proposed new step constructs a finite deterministic family of annular discounts using finite rectifiable graphs, codimension-two nonseparation, and stability of all minimizing paths. A Cameron–Martin count, using only individual shift norms and Hölder's inequality, closes the deterministic comparison interval. This is a proposed solution to an open research problem. The argument has been checked internally and is believed to be valid, but has not been independently verified. It should be read as a proposed research proof, not as an established result.

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1 The target and the construction inputs

Fix $d \geq 3$, $\xi > 0$, and the continuum LFPP distance exponent $Q = Q(\xi) > \sqrt{2d}$. The underlying field is the whole-space log-correlated Gaussian field, not the ordinary higher-dimensional Gaussian free field. Modulo constants its covariance is

$$\mathrm{Cov}((h,\phi),(h,\psi)) = \iint \phi(x)\psi(y)\log\frac{1}{|x-y|}\,dx\,dy, \qquad \int \phi = \int \psi = 0.$$

We choose $h_1(0) = 0$ when a representative is needed. Spherical averages are denoted by $h_r(z)$, and

$$s_r(z) = r^{\xi Q} e^{\xi h_r(z)}.$$

For an open set U, $D(\cdot,\cdot;U)$ is the internal metric. All metrics in the statement below are finite and induce the Euclidean topology. A metric assignment is a measurable map from distributions to such metrics.

Definition 1.1 (The four strong axioms)**.** The following requirements hold whenever the input is a whole-space LGF plus a possibly random continuous function.

(I) D_h is a length metric.

(II) For deterministic open U, $D_h(\cdot,\cdot;U)$ is measurable with respect to $h|_U$.

(III) Almost surely, simultaneously for all continuous f,

$$D_{h+f} = e^{\xi f}\cdot D_h,$$

where the right side minimizes the integral of $e^{\xi f}$ against D_h-arclength over paths with the prescribed endpoints.

(IV) For each fixed $r > 0$ and $z \in \mathbb{R}^d$, almost surely, simultaneously for all u, v,

$$D_h(ru+z, rv+z) = D_{h(r\cdot+z)+Q\log r}(u,v).$$

No separate assumption of scale tightness, rotation invariance, geodesic uniqueness, or confluence is made on a strong candidate.

Theorem 1.2 (Target statement)**.** *There exists an assignment satisfying the four strong axioms. Any two such assignments at the same (d,ξ,Q) differ almost surely by a deterministic positive constant, simultaneously on all endpoint pairs and after all continuous perturbations of the field.*

For the continuum mollifications covered by [1, Theorem 1.3], the normalized LFPP metrics converge in probability locally uniformly. Their limit is the assignment above, with its deterministic scalar fixed by the normalization of the approximation.

Scope of the approximation assertion. The existence input uses the explicit convolution kernels in [1, Remark 1.1], obtained from smooth compactly supported radial white-noise kernels. It does not establish tightness for every conceivable mollifier. The axiomatic characterization itself is independent of that choice of approximation. The parameter correspondence and kernel independence of $Q(\xi)$ are as in [1, Section 1.2], using [6].

The proof proceeds through an auxiliary class in which exact dilations are replaced by translation covariance and tightness across scales. This does not change the assumptions in Theorem 1.2; it permits the available subsequential limits to be compared before exact covariance has been proved.

Definition 1.3 (Auxiliary weak class)**.** Retain (I)–(III), require translation covariance, and require deterministic normalizers $\mathfrak{c}_r$ for which

$$\mathfrak{c}_r^{-1} e^{-\xi h_r(0)} D_h(r\cdot, r\cdot)$$

is tight locally uniformly over $0 < r < R$, for every finite R, with all limits inducing the Euclidean topology. For an LGF plus a bounded continuous function, require this for all $r > 0$. This is the weak class of [1, Section 1.2].

Theorem 1.4 (Auxiliary uniqueness statement)**.** *Any two assignments in this weak class with the same parameter ξ are deterministic positive multiples of one another.*

The main new argument is intended to prove Theorem 1.4. Once that has been proved, the final section derives Theorem 1.2, rather than assuming exact dilation covariance for the subsequential limits.

1.1 Imported results and exact uses

We use the following results of [1], whose construction builds on the tightness theorem of [2].

(E1) For every sequence of approximation parameters tending to zero, there is a deterministic subsequence converging in probability, locally uniformly, to a measurable weak metric assignment. The same subsequence statement holds for each LGF plus a bounded continuous function [1, Theorem 1.3].

(E2) The weak scale normalizers can be chosen as $\mathfrak{c}_r = r^{\xi Q}$ [1, Theorem 1.6].

(E3) The local metric comparison criterion [1, Proposition 3.13] applies to two jointly local, ξ-additive continuous length metrics: a sufficiently high uniform probability bound comparing the internal spherical diameter of one with the crossing distance of the other gives a deterministic global bi-Lipschitz comparison. In particular, two weak assignments admit such a comparison [1, Proposition 4.6].

(E4) For local events modulo constants on geometrically separated concentric annuli, arbitrarily high marginal probabilities imply exponential bounds for a prescribed positive fraction of failures. The probability threshold depends only on the desired exponential rate, fraction, and annular geometry [1, Lemma 2.7]. We use this for field-measurable internal metrics, so no additional metric randomness is present. A direct Gaussian proof of the part used here is also supplied in Lemma A.1.

(E5) For fixed connected open U and compact connected $K \Subset U$, the normalized internal diameter has uniformly bounded p-moments for

$$0 < p < \frac{2d}{\xi\gamma}, \qquad \gamma = Q - \sqrt{Q^2 - 2d} \in (0, \sqrt{2d}),$$

by [1, Proposition 4.11]. A Gaussian tilting argument below derives the required positive Holder exponent from this result. It does not use independence of spherical-average increments.

The input in (E3) applies because both internal metrics are functions of the restricted field; their joint locality is automatic. Applying locality after subtracting $h_r(z)$ and using Weyl scaling gives joint ξ-additivity.

For the two weak assignments write $D_h, \widetilde{D}_h$ and define

$$c_* = \sup\{c : \widetilde{D}_h \geq cD_h \text{ almost surely}\}, \qquad C_* = \inf\{C : \widetilde{D}_h \leq CD_h \text{ almost surely}\}.$$

All-pairs inequalities are equivalent to inequalities on a countable dense set, by continuity. Thus the endpoint constants are attained almost surely and

$$0 < c_* \leq C_* < \infty.$$

The same bounds apply to path lengths, internal metrics, and simultaneous continuous Weyl reweightings. The proof must show $c_* = C_*$.

2 Local compactness and quantitative continuity

We use the notation $B_r(z)$ for a Euclidean ball and $A_{a,b}(z) = B_b(z) \setminus \overline{B}_a(z)$. For a metric D, the length of a path is denoted by $\mathrm{len}(P; D)$. The inequalities $c_*D \leq \widetilde{D} \leq C_*D$ imply the same inequalities for lengths and internal distances. They also imply these inequalities after any common continuous positive Weyl weight.

Lemma 2.1 (Lengths under a continuous weight)**.** *Let D be a compatible length metric, let $w > 0$ be continuous, and let $w \cdot D$ be obtained by minimizing $\int w\, ds_D$ over D-rectifiable paths. This is a metric with the same topology as D. A continuous path on a compact parameter interval is rectifiable for one of the two metrics if and only if it is rectifiable for the other. For such a path P,*

$$\mathrm{len}(P; w \cdot D) = \int_P w\, ds_D.$$

The same assertion holds for a local metric coordinate with the short-distance realization property used below, when paths and the local comparisons lie in its inspected domain. Consequently common positive weights preserve metric comparison, and internal path lengths on a domain depend only on the restriction of the weight to that domain. More precisely, for open U,

$$(w \cdot D)(x, y; U) = \inf_{\substack{P: x \to y \\ P \subset U}} \int_P w\, ds_D,$$

where the infimum is over D-rectifiable paths.

Proof. First suppose that P is D-rectifiable. For each finite partition of P, using its own subpaths as competitors bounds the sum of weighted endpoint distances by $\int_P w\, ds_D$. This proves the upper bound for length.

For the reverse bound, fix a point of the compact image of P and a small neighborhood B on which the oscillation of w is at most δ. Take a smaller neighborhood B_0 compactly inside B. The D-distance from $\overline{B}_0$ to ∂B is positive. If two points of B_0 have D-distance sufficiently small, a path between them which stays in B costs at least $(\inf_B w)$ times their original distance, whereas a path leaving B already pays this much before its first exit. Thus, locally near the diagonal,

$$(w \cdot D)(x, y) \geq (\inf_B w) D(x, y).$$

This lower bound uses only a first-exit barrier and also applies to the local coordinates. Cover the image of P by finitely many such smaller neighborhoods and divide its parameter interval into pieces lying in one of them. On each piece, refine partitions sufficiently that consecutive endpoints satisfy the local comparison. Taking their supremum shows that its weighted-metric length is at least $(\inf_B w)$ times its original length. Summing the pieces gives a lower bound of $\int_P w\, ds_D - \delta \operatorname{len}(P; D)$. Let $\delta \downarrow 0$ to obtain equality.

For completeness, these local comparisons also give the topology and the converse rectifiability assertion, even when w has no global positive lower bound. Near a fixed point, short D-almost-minimizers stay in a neighborhood on which w is bounded above, yielding a local upper bound $(w \cdot D)(x, y) \le CD(x, y)$. A path leaving a smaller neighborhood pays a fixed positive weighted cost before its first exit. Together with the local lower comparison above, this proves nondegeneracy and equality of topologies.

Now let P be any continuous path on a compact parameter interval. Cover its compact image by finitely many of the smaller neighborhoods used for the lower comparison, and let $m_P > 0$ be the minimum of their positive lower weight bounds. Partition the parameter interval so that every piece stays in one of these neighborhoods. Refining partitions and summing the local lower comparisons gives

$$\operatorname{len}(P; w \cdot D) \ge m_P \operatorname{len}(P; D),$$

with infinite lengths allowed in this inequality. Thus finite weighted length forces finite original length, and the identity already proved then applies. This also shows that restricting to internal path metrics introduces no additional rectifiable paths after a field shift.

If $mD \le E \le MD$, the same inequalities hold for the lengths of every subpath, and hence for their arclength measures. Integration against the same positive continuous weight preserves these inequalities. Taking infima over paths gives $m(w \cdot D) \le w \cdot E \le M(w \cdot D)$. The assertion about internal lengths follows from the length identity on paths contained in that domain. □

2.1 Compact local configurations

Here are the deterministic compactness facts used to select finitely many perturbations. Domains in this subsection are deterministic.

Lemma 2.2 (Local convergence and internal metrics)**.** *Suppose continuous length metrics D_n on $\mathbb{R}^d$ converge locally uniformly to a metric D inducing the Euclidean topology. Then D is locally a length metric. For every connected open set U, the internal metrics $D_n(\cdot, \cdot; U)$ form a relatively compact family for local uniform convergence on $U \times U$. Every limit is finite, induces the Euclidean topology, is locally a length metric, and gives the same lengths as D to paths in U.*

More generally, these conclusions hold for a jointly converging family of internal metrics on nested domains, provided a common positive lower modulus and local upper moduli are retained. If $\overline{U_1} \Subset U$, a uniform bound $D_n(x_n, y_n; U_1) \le A$ supplies, in a joint limit, a path in $\overline{U_1}$ of length at most A for the U coordinate.

Proof. Fix a compact ball K contained in the domain of inspection and a slightly larger compact ball K'. Distances, for D, between K and $\partial K'$ have a positive minimum. The same is true, with a smaller uniform constant, for D_n. Consequently a short D_n-almost-minimizer starting in K cannot leave K'. Parameterize it by D_n-length, extending constantly to a common bounded time interval. Local uniform convergence and the positive lower modulus give Euclidean equicontinuity. Arzelà–Ascoli and lower semicontinuity of length yield short limiting paths in K'. In particular D is locally a length metric.

Every connected open subset of a space with this local length property is rectifiably connected. Indeed, the class reachable from a fixed point by finite-length paths is open, as is its complement. Connectedness makes the class the whole set. Fix a finite-length path in U for D. Divide it into finitely many pieces sufficiently short that the preceding confinement argument applies in balls compactly contained in U. Replacing the pieces by D_n-almost-minimizers bounds their internal distances. A finite covering of any prescribed compact set, together with finitely many such connecting paths, gives uniform bounds for the internal metrics on that compact set. The same short-path argument gives a common upper modulus there. Arzelà–Ascoli on a countable compact exhaustion proves relative compactness. Internal metrics dominate D_n, so every limit has a positive lower modulus. There is no loss of topology through escape to a domain boundary: for a fixed small ball B about x, every internal competitor from x to a point outside B must pay the distance from x to ∂B. This has a uniform positive lower bound and passes to the limit for every endpoint outside B. Near the diagonal, a short global competitor cannot exit U; hence global and internal distances agree there. Their limits consequently give identical path lengths in U.

For the last assertion take U_1-competitors of length at most $A+1/n$. Their images lie in the fixed compact set $\overline{U_1} \Subset U$. Reparameterization, the common inverse modulus, and lower semicontinuity give the required path. It may meet ∂U_1, but it remains strictly inside U. The same near-diagonal argument shows equality of the coordinate path lengths on overlapping domains. □

Applying this lemma after a subsequential coupling supplied by tightness shows that normalized internal metrics on any fixed finite list of connected domains have jointly tight laws, uniformly over scale. We use the two domains

$$U = A_{1/2,5}, \qquad U_1 = A_{3/4,7/2}, \qquad \overline{U_1} \Subset U.$$

A local configuration consists of the two normalized internal metrics on each of these domains. By imposing common upper bounds and moduli on a countable exhaustion, and common positive lower moduli on Euclidean-separated compact pairs, we obtain compact sets $\mathcal{C}$ of configurations of arbitrarily high probability, uniformly over scale. We take their closures in the product local uniform topology. The proof above shows that the closure retains the local length and compatibility properties just stated, the comparison inequalities, and the path-length agreement of the coordinates. Later open neighborhoods are relative to such a fixed $\mathcal{C}$.

Lemma 2.3 (Uniform local realization of distances)**.** *For the compact family $\mathcal{C}$ just constructed, fix $K \Subset O \Subset U$, where K is compact and O is open. There is $\delta > 0$ such that, for every configuration in $\mathcal{C}$ and either metric coordinate D^U, any $x, y \in K$ with $D^U(x,y) < \delta$ are joined by a path in $\overline{O}$ whose D^U-length is exactly $D^U(x,y)$. The analogous assertion holds inside U_1.*

Proof. The uniform lower modulus on compact sets gives $a > 0$ such that every original coordinate used to construct $\mathcal{C}$ has distance at least a between K and ∂O. Here these are disjoint compact sets with positive Euclidean separation, and $\overline{O} \Subset U$. Take $\delta = a/4$.

Let D_n^U be original internal length metrics converging to a specified coordinate D^U in the compact closure. For fixed $x, y \in K$ with $D^U(x,y) < \delta$, take D_n^U-competitors of length at most $D_n^U(x,y) + 1/n$. For large n their length is less than $a/2$, so none can exit O: a first exit would already cost at least a. Parameterize them at constant speed on $[0,1]$. The common inverse modulus on $\overline{O}$ gives Euclidean equicontinuity. A uniformly convergent subsequence has image in $\overline{O}$, joins x to y, and has D^U-length at most $D^U(x,y)$ by lower semicontinuity. The reverse inequality holds for every path in a metric space. Thus the limit realizes the distance. Original configurations can be treated by a constant sequence in the same argument. The choice of δ is uniform over the compact family and the two metric types. □

Lemma 2.4 (Quantitative recovery of a path)**.** *Fix $K \Subset O \Subset U$, with K compact and O open, and a positive continuous weight w. Let D, D' be either of the corresponding metric coordinates of two configurations in $\mathcal{C}$, and put*

$$\epsilon = \sup_{x,y\in\overline{O}} |D'(x,y) - D(x,y)|, \qquad M = \max_{\overline{O}} w.$$

There is a common nondecreasing inverse modulus χ, with $\chi(s) \to 0$ as $s \downarrow 0$, such that $D''(x,y) \le s$ implies $|x-y| \le \chi(s)$ for every coordinate D'' in $\mathcal{C}$ and $x, y \in \overline{O}$. Write

$$\omega_w(r) = \sup_{\substack{x,y\in\overline{O}\\ |x-y|\le r}} |w(x) - w(y)|.$$

Let $\delta_{\mathrm{loc}} > 0$ be the threshold in Lemma 2.3 for K, O. If $\epsilon \le \delta$ and $2\delta < \delta_{\mathrm{loc}}$, then every D-rectifiable path $P \subset K$ of length at most L has a D'-rectifiable replacement $P' \subset \overline{O}$ with the same endpoints and

$$\mathrm{len}(P'; D') \le L + (L/\delta + 1)\epsilon, \tag{1}$$

$$\int_{P'} w\, ds_{D'} \le \int_P w\, ds_D + 2L\omega_w(\chi(2\delta)) + 2M(L/\delta + 1)\epsilon. \tag{2}$$

The replacement can be parameterized on the same interval as P so that their pointwise Euclidean distance is at most $2\chi(2\delta)$.

Proof. The common positive lower modulus on $\overline{O}$ gives χ: take the supremum of the Euclidean separations of all pairs at coordinate distance at most s. If this supremum failed to tend to zero, compactness of $\mathcal{C}$ and $\overline{O}$ would give distinct points at zero distance for one limiting coordinate.

A constant path needs no replacement. Otherwise divide P by its D-arclength into $N \le L/\delta + 1$ consecutive pieces of lengths $\ell_i \le \delta$, with successive endpoints z_i, z_{i+1}. For every piece,

$$D'(z_i, z_{i+1}) \le D(z_i, z_{i+1}) + \epsilon \le \ell_i + \epsilon \le 2\delta.$$

Uniform local realization supplies a D'-geodesic P_i' between these endpoints in $\overline{O}$. Concatenate these paths. Summing their lengths proves (1).

Every point of P_i' is within Euclidean distance $\chi(2\delta)$ of z_i, since its D'-distance from z_i is at most the total length of P_i'. The same holds for the original piece, using its D-length bound δ. Thus, with $\Omega = \omega_w(\chi(2\delta))$,

$$\int_{P_i'} w\, ds_{D'} \le (w(z_i) + \Omega)(\ell_i + \epsilon), \qquad \int_{P|_i} w\, ds_D \ge (w(z_i) - \Omega)\ell_i.$$

Subtract and sum. Since $w(z_i) + \Omega \le 2M$, the result is (2). Parameterize each new piece on the parameter interval of its original piece. Both points then lie within $\chi(2\delta)$ of the same z_i, proving the last assertion. Constant portions of the original path can simply be retained as constant portions. □

In particular, if $D_n \to D$ on $\overline{O} \times \overline{O}$, take $\delta_n = \sqrt{\epsilon_n} + 1/n$. Every fixed rectifiable path in K then has uniformly convergent replacements whose weighted lengths have upper limit at most its weighted length. The error estimate is uniform over all such paths of bounded length. Moving endpoints are handled by adding short locally realized connections; the common upper modulus makes their weighted costs tend to zero. This is the localized path-recovery argument underlying fixed-weight continuity, comparable to [8, Lemma 7.1]. Here it is proved for our local metric coordinates and with explicit confinement.

Lemma 2.5 (Continuity under a fixed weight)**.** *Let local metric configurations converge in $\mathcal{C}$, and fix a positive continuous weight w. Suppose the endpoints and all relevant minimizing paths and recovering competitors lie in a common compact subset of the inspected domain. Then the weighted distances converge, uniformly for the endpoint pairs under consideration. Limits of weighted minimizing paths, parameterized by length and with uniformly bounded lengths, are minimizing paths for the limiting local length structure.*

Proof. On the confining compact set, w has positive minimum. Weighted length bounds thus bound the unweighted lengths. The common lower modulus gives compactness of the paths. Subdivide into pieces on which w oscillates by at most δ; ordinary lower semicontinuity of unweighted length on each piece gives the weighted lower bound as $\delta \downarrow 0$. For the upper bound apply Lemma 2.4 to a limiting rectifiable competitor, in a slightly larger compact neighborhood of its image. Its explicit error tends to zero. Short connections handle moving endpoints, as in the observation after that lemma. A subsequence argument then makes the distance convergence uniform. Equality of the upper and lower bounds identifies limits of minimizing paths. □

2.2 A positive Hölder exponent without radial independence

We need a quantitative upper modulus only for the final discretization of endpoints. The following derivation uses the diameter moment input, rather than independent spherical increments.

Lemma 2.6 (Uniform upper modulus)**.** *For every $0 < \alpha < \xi(Q - \sqrt{2d})$ and $L < \infty$, there is $a > 0$ such that*

$$\mathbb{P}\left[\sup_{\substack{x,y\in B_{LR}\\ |x-y|\le \varepsilon R}} D_h(x,y) > \varepsilon^{\alpha} s_R(0)\right] = O(\varepsilon^{a}),$$

uniformly in $R > 0$. The same argument gives upper bounds for internal diameters of balls inside a fixed enlarged macro ball.

Proof. Set $t = \varepsilon R$, fix $z \in B_{(L+2)R}$, and write

$$X = h_{2t}(z) - h_R(0), \qquad H = h - h_{2t}(z).$$

Let σ and τ be uniform probability measures on $\partial B_{2t}(z)$ and $\partial B_R(0)$, respectively. If $F_\mu(w) = \int \log|w-u|^{-1}\,\mu(du)$, then

$$k(w) := \operatorname{Cov}(X, H(w)) = F_\sigma(w) - F_\tau(w) - \int (F_\sigma - F_\tau)\,d\sigma.$$

This is an identity for the covariance with the restricted generalized field. The function k is continuous. Moreover

$$\operatorname{Var} X = \log(R/t) + O_L(1), \qquad \sup_{B_{2t}(z)} |k| = O_L(1).$$

For the second estimate, the two small-sphere terms have their $\log(1/t)$ terms cancel. Their remaining unit-sphere potentials are bounded on a fixed compact set. The large-sphere terms have the same property after scaling by R. The logarithmic potential of the unit sphere is continuous even on the sphere: the integral over a spherical cap of radius δ is bounded by $O(\delta^{d-1}(1 + |\log\delta|))$, uniformly in its center. The variance estimate follows from the same covariance formula.

Put

$$M(H) = (2t)^{-\xi Q} \sup_{u,v \in \overline{B}_t(z)} D_H(u, v; B_{2t}(z)).$$

Locality makes this a functional of the restricted field. The diameter input gives $\mathbb{E}[M(H)^p] \le C_p$, uniformly in z, t, for $0 < p < 2d/(\xi\gamma)$. Gaussian tilting, followed by the local Weyl comparison, yields

$$\begin{aligned}\mathbb{E}[e^{\xi p X} M(H)^p] &= e^{\xi^2 p^2 \operatorname{Var}(X)/2} \mathbb{E}[M(H + \xi p k)^p] \\ &\le C_{p,L} e^{\xi^2 p^2 \operatorname{Var}(X)/2}.\end{aligned}$$

The identity first holds for bounded measurable truncations and then follows in this form by monotone convergence. Weyl scaling is applicable to the continuous deterministic function k; only its bound on the inspected ball enters the estimate. It follows that

$$\mathbb{E}\left[\left(s_R(0)^{-1} \sup_{u,v \in \overline{B}_t(z)} D_h(u, v; B_{2t}(z))\right)^p\right] \le C_{p,L} \varepsilon^{\xi Q p - \xi^2 p^2/2}.$$

For $s \in (0, \xi Q)$ choose $p = (\xi Q - s)/\xi^2$. This is an admissible moment, because $Q/\xi < 2d/(\xi\gamma)$ is equivalent to $\gamma^2 < 2d$. Markov's inequality gives the exponent $(\xi Q - s)^2/(2\xi^2)$ for the event that this diameter exceeds $\varepsilon^s s_R(0)$.

Choose $\alpha < s < \xi(Q - \sqrt{2d})$, so this exponent is strictly larger than d. At each scale $2^{-j}\varepsilon R$, use a grid of spacing $2^{-j}\varepsilon R/(4d)$ in the enlarged macro ball. A union bound over its $O((2^{-j}\varepsilon)^{-d})$ points and then over $j \ge 0$ has a positive polynomial rate. Every pair at distance at most εR lies in one of these grid balls at a comparable scale. Lowering s to α absorbs the fixed constants. This proves the asserted modulus. □

2.3 Properness and measurable geodesics

Lemma 2.7 (Properness and confinement)**.** *For a whole-space LGF, D_h and $\widetilde{D}_h$ are almost surely proper. For every $L < \infty$ and $\delta > 0$ there is $A < \infty$ such that, uniformly in R, with probability at least $1 - \delta$, all geodesics for either metric with endpoints in B_{LR} remain in B_{AR}.*

Proof. Normalized diameters of B_{LR} are tight. The crossing distance of $A_{tR/2,tR}$, divided by $s_R(0)$, is

$$(t/2)^{\xi Q} e^{\xi(h_{tR/2}(0) - h_R(0))} Z_{tR/2},$$

where Z_r is bounded away from zero in probability, uniformly in r. The Gaussian increment has variance at most $\log t + O(1)$. Since $Q > 0$, the displayed expression tends to infinity in probability uniformly in R; no independence of its factors is needed. A path leaving a sufficiently large ball must pay this crossing cost. This proves the confinement statement. Choose a sufficiently fast deterministic sequence $t_k \to \infty$ so that the probability of a crossing cost less than k, with $R = 1$, is summable. Borel–Cantelli shows that metric distances from a fixed Euclidean compact set to ∂B_{t_k} tend to infinity. Thus closed metric balls are Euclidean bounded and closed, hence compact. A proper length space is geodesic. Comparison gives the same assertions for the other metric. □

Fix a measurable selector of constant-speed geodesics which is unchanged when the entire metric is multiplied by a scalar. Here is one precise construction. For each integer n, consider the paths in $\overline{B}_n$ joining the endpoints and satisfying

$$D(P(s), P(t)) = |s - t| D(x, y), \qquad s, t \in [0, 1] \cap \mathbb{Q}.$$

Their set is compact, by the inverse modulus, and its graph is Borel. Compact-section measurable selection makes the nonempty fibers a measurable compact-valued map. Choose the least n with a nonempty fiber and apply a fixed selector on compact subsets of continuous-path space. Both the fiber and the least n are unchanged by scalar multiplication of D. Properness makes some fiber nonempty. Denote the selected path by P_h. No uniqueness or continuity of this selector is required.

All local witness events below are measurable. For example, internal distance sublevel events are projections of sets of paths confined to a compact exhaustion and with bounded Lipschitz constant; their fibers are compact. Geodesic witnesses with strict endpoint inequalities are countable unions of such closed compact-section conditions. The same selection argument applies. Excursion components are enumerated by rational times; first and last visits to a fixed closed collector are measurable hitting times. Universal statements about geodesics will be consequences of deterministic lemmas, not definitions by an uncountable intersection of events.

3 Collapse onto a finite rectifiable graph

A finite rectifiable graph in a metric space is a compact embedded finite topological graph whose edges are simple rectifiable arcs. Giving each edge its ambient metric length defines its intrinsic metric d_K. The graph is compact for d_K, and this metric has the inherited topology. Ambient lengths of paths in K equal their intrinsic lengths.

Lemma 3.1 (Collapse with weighted attraction)**.** *Let (X, D) be a proper length space and $K \subset X$ a connected finite rectifiable graph. Let $\lambda_n \downarrow 0$, $\eta_n/\lambda_n \to 0$, and let continuous weights satisfy*

$$\lambda_n \le w_n \le 1, \qquad w_n|_K = \lambda_n, \qquad w_n \ge a_0 > 0 \quad \text{off } N^D_{\eta_n}(K).$$

Set $D_n = w_n \cdot D$. For x_n, y_n in the closed η_n-tube, choose $u_n, v_n \in K$ at D-distance at most η_n from them. If $u_n \to u$ and $v_n \to v$, then

$$\lambda_n^{-1} D_n(x_n, y_n) \longrightarrow d_K(u, v).$$

Every sequence of D_n-geodesics between these endpoints, parameterized with D_n-speed λ_n and extended constantly to a common interval, has a subsequence converging uniformly in D to a unit-speed d_K-geodesic P from u to v, with the same constant extension. Along that subsequence,

$$\sup_t \lambda_n^{-1} D_n(P_n(t), P(t)) \longrightarrow 0. \tag{3}$$

The same conclusions hold for local length structures when all paths and competitors used here are confined to a fixed compact subset of their domain.

Proof. Access paths and a shortest graph path give the recovery bound

$$L_n := \lambda_n^{-1} D_n(x_n, y_n) \le d_K(u_n, v_n) + 2\eta_n/\lambda_n. \tag{4}$$

The path P_n has D-speed at most one. Its total D-length off the tube is at most $\lambda_n L_n/a_0$, so its maximum D-distance from K tends to zero. The lengths L_n are bounded. Properness, or the stipulated local confinement, and Arzelà–Ascoli yield a uniform limit P in K. Lower semicontinuity and (4) imply

$$d_K(u, v) \le \operatorname{len}(P; D) \le \liminf_n \operatorname{len}(P_n; D) \le \liminf_n L_n \le \limsup_n L_n \le d_K(u, v).$$

Thus every inequality is an equality. In particular L_n and $\mathrm{len}(P_n; D)$ both tend to $d_K(u, v)$, and the one-Lipschitz limit attains the distance in precisely that time. It is the claimed graph geodesic.

Let T_n be the amount of the normalized weighted time spent off the tube. By Lemma 2.1, the arclength measures satisfy $d\ell_{D_n} = w_n\, d\ell_D$. In the chosen parameterization $d\ell_{D_n} = \lambda_n\, dt$ before the constant extension. Therefore the original length measure is at most $(\lambda_n/a_0)dt$ off the tube and at most dt elsewhere. Hence

$$\mathrm{len}(P_n; D) \le L_n - (1 - \lambda_n/a_0)T_n,$$

so $T_n \to 0$. From any $P_n(t)$ outside the tube, follow its excursion to the tube, at D_n-cost at most $\lambda_n T_n$. Then take an access path to a point $q_n(t) \in K$, at cost at most $\eta_n + o(\eta_n)$. Uniform convergence gives $\sup_t D(q_n(t), P(t)) \to 0$. On the compact graph the inverse of the identity from (K, d_K) to (K, D) is uniformly continuous, so also $\sup_t d_K(q_n(t), P(t)) \to 0$. Travel in K, where the weight is λ_n. Dividing the resulting bound by λ_n proves (3). □

Lemma 3.2 (Transfer of a graph shortcut)**.** *Suppose $mD \le E \le MD$ and a finite collection of marked arcs $\sigma_i \subset K$ has endpoints u_i, v_i satisfying*

$$D(u_i, v_i) \ge a_1 > 0, \qquad \mathrm{len}(\sigma_i; E) \le c_0 D(u_i, v_i), \qquad c_0 < c_1 < M.$$

Assume every graph geodesic with endpoint D-distance at least b traverses one complete marked arc. For all sufficiently large n, every D_n-geodesic between points of the closed η_n tube with endpoint D-distance at least b contains a subsegment of D_n-length at least $\lambda_n a_1/2$ whose endpoints are joined by a path of $w_n \cdot E$-length at most c_1 times that length. In the local version this shortcut stays in the same confining domain.

Proof. Otherwise choose a sequence of failing geodesics. The collapse lemma gives a limiting graph geodesic with separated endpoints, which traverses some σ_i. At its entrance and exit times, the corresponding approximating points are at D_n-distance $o(\lambda_n)$ from u_i, v_i. Use the actual connection paths constructed in the collapse proof, rather than an infimum over unrelated comparison paths. Lemma 2.1 bounds their weighted E-lengths by M times their weighted D-lengths, hence by $o(\lambda_n)$. Concatenating them with the marked arc gives a concrete shortcut, whose middle part has cost exactly $\lambda_n \mathrm{len}(\sigma_i; E)$. In the local version all these paths lie in the confining domain, so the same estimate is an internal-distance bound there. The intervening D_n-length divided by λ_n converges to the graph length of this subarc, at least $D(u_i, v_i)$. The strict gap $c_0 < c_1$ contradicts failure. This contradiction allows arbitrary endpoints and arbitrary minimizing paths, which is the asserted uniformity. □

4 Routing the graph in dimension at least three

We use the classical codimension-two nonseparation theorem: a relatively closed subset of covering dimension at most $d - 2$ cannot disconnect a connected Euclidean region in dimension d; see the theorem recalled in [5]. A finite closed union of simple arcs has covering dimension at most one. This topological dimension statement applies even when the arcs have large Euclidean Hausdorff dimension.

Lemma 4.1 (An embedded graph with unavoidable marked arcs)**.** *Fix a sufficiently fine finite simplicial mesh of ∂B_2 and a spanning tree of its edge graph. Give its vertices disjoint small neighborhoods and its edges thin tubes, disjoint away from the incident vertex neighborhoods. For*

each edge choose a central compact box, disjoint from every other tube, and inside it a simple rectifiable arc γ_e satisfying

$$D(\gamma_e(0),\gamma_e(1)) \geq a_1, \quad \operatorname{len}(\gamma_e; D) \leq A_1, \quad \operatorname{len}(\gamma_e; E) \leq c_2 D(\gamma_e(0),\gamma_e(1)),$$

where $c_2 < c_3$. All regions lie in a fixed thin open collector V around ∂B_2. Then one can join the mesh vertices and these arcs into a connected finite rectifiable graph $K \subset V$, with a marked middle subarc of each γ_e, whose endpoint distance is at least $a_1/2$ and whose E-length is at most c_3 times that distance.

After removing the relative interiors of the marked arcs, each component belongs to a single mesh vertex and its incident stubs. If each enlarged vertex star, including its incident edge tubes, has D-diameter less than $b/8$, every graph geodesic with endpoint distance at least $b/4$ traverses a complete marked arc.

Proof. Parameterize each γ_e by D-arclength on $[0, L_e]$. Choose $\tau > 0$ so small that

$$2\tau < (1 - c_2/c_3)a_1, \qquad 2\tau < a_1/2.$$

Fix the marked arc $\sigma_e = \gamma_e([\tau, L_e - \tau])$ before routing any connector. The two permitted attachment pieces are $\gamma_e([0,\tau))$ and $\gamma_e((L_e - \tau, L_e])$. Trimming at most τ of D-length from each end gives

$$D(\sigma_e(0),\sigma_e(1)) \geq a_1 - 2\tau > a_1/2, \qquad \operatorname{len}(\sigma_e; E) \leq c_2 a_e < c_3(a_e - 2\tau),$$

where $a_e = D(\gamma_e(0),\gamma_e(L_e)) \geq a_1$. Thus the marked arc has the required ratio.

We distinguish a geometric enlarged star from an attachment component of the partially constructed graph. The former is an open neighborhood containing one vertex neighborhood and all its incident tubes. The latter is the compact component containing that mesh vertex after the relative interiors of all marked arcs have been removed. Initially the graph consists of the isolated mesh vertices and all γ_e. Each vertex attachment component is then a singleton; the unconnected arc stubs are separate components.

Process the tree edges successively. Maintain the following invariant: the union constructed so far is a finite embedded rectifiable graph; every vertex attachment component contains exactly one mesh vertex and lies in its geometric enlarged star; and no connector meets the interior of a marked arc. The central boxes are disjoint from every other edge tube. Routing regions for an edge consist of its corresponding tube half, its whole central box, and the corresponding vertex neighborhood; they exclude the opposite vertex neighborhood and every other central box. Previously processed tubes can meet such a region only in its designated vertex neighborhood.

For the left connector of an edge e, remove $\gamma_e([\tau, L_e])$ from its left routing region. This is a relatively closed subset of covering dimension at most one. The remaining open region is connected by the codimension-two nonseparation theorem and is rectifiably connected by the local length property. Join the left mesh vertex to $\gamma_e(0)$ by a finite-length path P there. Let t be its first contact with γ_e, and let $s < t$ be its last preceding contact with the left attachment component. Both times exist, since these are compact disjoint sets containing the initial and terminal points of the relevant portion of P. The segment $P|_{[s,t]}$ meets neither set in its interior, and its terminal point is in $\gamma_e([0,\tau))$.

Replace this segment by a shortest D-path inside its compact image. To justify this operation, the original segment gives a finite upper length bound; constant-speed parameterizations of a minimizing sequence are equicontinuous and take values in that compact image. A limiting minimizer exists by lower semicontinuity, and is injective because its endpoints differ. Its interior still avoids

both attachment sets. The new arc therefore adds only its two endpoint attachments to the existing finite graph. All other existing pieces are excluded by the routing-region geometry.

For the right connector, remove from the right routing region the first connector and $\gamma_e([0, L_e - \tau])$. Their union is relatively closed and has covering dimension at most one; it contains neither the right mesh vertex nor $\gamma_e(L_e)$. Repeat the preceding construction, using the right attachment component and first contact with γ_e. This connector meets γ_e in its permitted terminal piece and avoids the first connector and the marked arc. Last-contact trimming again prevents any interior intersection with the existing right component. The invariant is preserved: on deleting the marked interiors, the two new attachments connect only the appropriate unconnected stub to its vertex component.

At completion every stub has been attached, and the original spanning tree ensures that the full graph K is connected. Deleting the marked interiors leaves precisely the vertex attachment components. If a simple path traversing no marked arc in full meets no such component, it lies inside one marked arc, hence inside either incident geometric enlarged star. Otherwise it can meet only one attachment component: reaching another would require traversing a whole marked arc. Its initial and terminal portions can lie in partial marked arcs incident to that component. Thus in either case both endpoints lie in one geometric enlarged star, including its incident tubes, and have D-distance less than $b/8$. Every graph geodesic is simple, so endpoint separation at least $b/4$ forces a complete marked-arc traversal. □

Lemma 4.2 (A smooth trap for every substantial passage)**.** *Suppose the graph above lies in $V \Subset A_{1,3}$ and every point of $\overline{V}$ has an access path to a mesh vertex of D-length at most ε_0. Suppose fixed collar barriers confine sufficiently short paths between collector points to $A_{1,3}$. Fix $c_3 < c' < M$. If ε_0 is sufficiently small compared with b and the collar bounds, there is a nonnegative $f \in C_c^\infty(A_{1,3})$, positive exactly on V, with these properties:*

(i) *Any two collector points can be connected at $e^{-\xi f} \cdot D$ cost less than $3\varepsilon_0$.*

(ii) *Every minimizing path between collector points whose original endpoint D-distance is at least b contains a subsegment of weighted length at least $c > 0$ whose endpoints are joined inside $A_{1,3}$ by a path of $e^{-\xi f} \cdot E$-length at most c' times that length.*

The construction can first enforce (ii) with threshold $b/2$, length $2c$, and ratio strictly below c'. With these margins, the final assertions persist on a relative neighborhood of the configuration in a compact family $\mathcal{C}$ as above.

Proof. Choose smooth $0 \le f_0 \le 1$, positive exactly on V. Let $\lambda_n \downarrow 0$, $\eta_n = o(\lambda_n)$, and choose smooth ϕ_n equal to one near K, supported in $N_{\eta_n}^D(K) \Subset V$, with $0 \le \phi_n \le 1$. For $M_n = \xi^{-1} \log \lambda_n^{-1} > 1$ set

$$f_n = f_0 + (M_n - f_0)\phi_n, \qquad w_n = e^{-\xi f_n}.$$

These weights satisfy the collapse hypotheses with $a_0 = e^{-\xi}$. Access and graph paths have total cost at most $2\varepsilon_0 + \lambda_n \operatorname{diam}(K, d_K)$. Take $\varepsilon_0 < e^{-\xi} b/200$, with additional smallness relative to the collar bounds, and then n large. This gives (i) and confines every minimizing segment between collector points to the fixed compact annulus.

If the original endpoint distance is at least $b/2$, such a minimizer must hit the tube. Otherwise its cost is at least $a_0 b/2 > 3\varepsilon_0$. The sum of its original lengths before its first tube hit and after its last tube hit is at most $a_0^{-1} 3\varepsilon_0 < b/8$. Its first and last tube points therefore have distance at least $3b/8$. Apply the graph shortcut lemma, with threshold $b/4$ and a ratio strictly between c_3 and c', to the minimizing middle segment. Choose one sufficiently large n, and then a smaller positive c, to reserve the stated length and ratio margins.

For stability, keep this $f = f_n$ fixed, and write $w = e^{-\xi f}$. Let (D_k, E_k) be a converging sequence of configurations in $\mathcal{C}$ with limit (D, E). We first establish confinement without presupposing continuity of weighted minimizing paths. For every pair of collector endpoints, choose the original access-and-graph competitor. Its original D-length is at most

$$L_0 = 2\varepsilon_0 + \operatorname{diam}(K, d_K),$$

and its weighted cost is less than $3\varepsilon_0$. All these competitors lie in one compact subset K_0 of $A_{1,3}$. To see the uniform buffer for the access pieces, each has original length at most ε_0 from a collector point. If a point of such a piece approached $\partial A_{1,3}$, continuity of the local metric on the larger inspected compact set would make its distance to that boundary tend to zero. The triangle inequality would contradict the original collar bound, which exceeds ε_0. The graph itself is already compactly contained in V.

Choose $K_0 \Subset O \Subset A_{1,3}$ and apply (2) with $L = L_0$ and $\epsilon_k = \sup_{\overline{O}\times\overline{O}} |D_k - D|$. Its error tends to zero uniformly over all the chosen competitors. Hence, for large k, every pair of collector points has a D_k-weighted competitor of cost less than $4\varepsilon_0$. This proves the required uniform connection bound directly, before any compactness assertion about weighted minimizing paths.

The collar distances have a positive margin and converge on fixed compact sets. A path approaching either boundary of $A_{1,3}$, after its last collector visit, pays an unaffected cost bounded below uniformly in k. The common upper modulus near those boundaries leaves a fixed positive Euclidean buffer between every sufficiently short minimizing path and the boundaries. Consequently all the minimizing paths under consideration lie in one compact subannulus. Weighted-distance continuity for $w \cdot D_k$ from Lemma 2.5 is therefore now applicable. The E_k comparison will be obtained by recovering the concrete local shortcut constructed in Lemma 3.2.

Suppose the final trap were to fail along this sequence. Take a failing minimizing path Q_k between collector endpoints with original D_k-distance at least b, parameterized on $[0, 1]$ with constant weighted speed L_k. The fixed positive minimum of w bounds its original speed. The common inverse modulus and confinement give a subsequence converging uniformly to Q, with endpoints x, y. Weighted-distance continuity gives $L_k \to L$, and Q is a constant-speed weighted minimizer. Its original endpoint distance is at least b. The local coordinate can be used in this threshold: any competitor of original cost below b is confined by the unweighted collars, so global and inspected internal threshold tests agree.

The stronger construction for the limiting configuration gives times $u < v$, a ratio $\widehat{c} < c'$, and a concrete shortcut S joining $Q(u)$ to $Q(v)$ inside the inspected annulus, with

$$(v - u)L \geq 2c, \qquad \operatorname{len}(S; w \cdot E) \leq \widehat{c}\,(v - u)L.$$

The image of S is compactly contained in the annulus. Apply Lemma 2.4 to E, E_k on a compact neighborhood of this image. The original E-length of S is finite because the fixed weight has positive minimum there. The error in (2) tends to zero. Include short connections to the moving endpoints $Q_k(u), Q_k(v)$. This gives paths S_k inside the same inspected annulus such that

$$\limsup_k \operatorname{len}(S_k; w \cdot E_k) \leq \operatorname{len}(S; w \cdot E).$$

For large k, the original Q_k subsegment has weighted length $(v - u)L_k \geq c$, and S_k has length less than $c'(v - u)L_k$. This contradicts failure of the final trap, for any choice of the minimizing path and endpoints.

The connection costs between all collector points persist uniformly by the same upper recovery argument, with their reserved strict margin. Thus all the final assertions hold on a relative neighborhood of the configuration in $\mathcal{C}$. No openness assertion for arbitrary unrelated metric pairs is needed. □

5 Regular extreme ratios occur at many scales

We prove a continuous-metric version of the scale-frequency step. Let (A_h, B_h) be either $(D_h, \widetilde{D}_h)$ or $(\widetilde{D}_h, D_h)$, and suppose

$$lA_h \le B_h \le UA_h, \qquad 0 < l \le U < \infty.$$

Fix the absolute scale lattice $\mathcal{S} = \{64^j : j \in \mathbb{Z}\}$.

5.1 Regular witnesses

Choose a small failure probability $\delta > 0$, specified below. Uniform scale tightness, internal diameter tightness and the local modulus give constants

$$0 < a < A_0 < \infty, \qquad 0 < \ell < a/100, \qquad 0 < \beta_0 < 1,$$

and local events $F_r(z)$ of probability at least $1 - \delta$ with these properties, in units of $s_r(z)$:

(R1) The A_h-distance across $A_{r,2r}(z)$ is at least a, and the internal A_h-diameter of $\overline{A}_{r,2r}(z)$ in $A_{3r/4,3r}(z)$ is at most A_0.

(R2) From $\overline{A}_{r,2r}(z)$ to either boundary of $A_{3r/4,3r}(z)$, the distance for each metric, measured internally in the larger inspected annulus $A_{r/2,5r}(z)$, is greater than $4\max(1, U)\ell$.

(R3) Points of $\overline{A}_{r,2r}(z)$ at Euclidean distance less than $\beta_0 r$ have internal A_h-distance less than $\ell/4$ in $A_{3r/4,3r}(z)$.

To obtain these choices in order, first choose positive collar and crossing bounds and a finite diameter bound. Next choose ℓ small enough for (R2), and then choose β_0 using the upper modulus. Decrease a or increase A_0 if necessary. All conditions are local modulo constants.

For $t \in (0, U)$ let $H_r(t)$ be the event $F_r(0)$ together with the existence of a path P in $\overline{A}_{r,2r}(0)$ which is an internal A_h-geodesic in $A_{3r/4,3r}(0)$ and satisfies

$$\ell/2 \le s_r(0)^{-1} \operatorname{len}(P; A_h) \le \ell, \qquad B_h(P(0), P(1); A_{3r/4,3r}) > t \operatorname{len}(P; A_h). \tag{5}$$

The collar bounds imply that both endpoint distances in (5) agree with their global versions. Also $|P(0) - P(1)| \ge \beta_0 r$. Translating gives $H_{z,r}(t)$ with the same probability.

Lemma 5.1 (Absence of a witness gives a crossing improvement)**.** *On $F_r(z) \setminus H_{z,r}(t)$, every global A_h-geodesic subpath crossing $A_{r,2r}(z)$ while staying in its closure has endpoints at B_h-distance at most t times its A_h-length.*

Proof. Its length L lies in $[a, A_0]s_r(z)$. Divide it into $\lceil L/(\ell s_r(z))\rceil$ equal-length pieces. Every piece has length in $[\ell/2, \ell]s_r(z)$ and is an internal geodesic. Absence of (5) gives the required B_h bound for each piece. Add these bounds. □

Lemma 5.2 (Rounded-center shell covering)**.** *Fix $\rho \in (0, 1)$ and $L < \infty$. Suppose a deterministic subset $\mathcal{T} \subset \mathcal{S} \cap [\varepsilon^2 R, \varepsilon R]$ contains at least $\rho \log_{64} \varepsilon^{-1} - O(1)$ scales. At each scale suppose local events $E_{z,r}$, determined modulo constants by the field in $A_{r/2,5r}(z)$, have probability at least p, uniformly in z, r. For p sufficiently close to one, with polynomially high probability, every point of B_{LR} is contained in a ball $B_{r/2}(z)$ with*

$$r \in \mathcal{T}, \qquad z \in \frac{r}{100d}\mathbb{Z}^d, \qquad E_{z,r} \text{ occurring.}$$

The power of the exceptional probability can be prescribed by choosing p closer to one. This choice of p is independent of L and of the additive constant in the lower bound for $|\mathcal{T}|$; those quantities affect only the prefactor and the threshold for ε. All rates are uniform in R.

Proof. For each point x of the grid $(\varepsilon^2 R/(1000d))\mathbb{Z}^d \cap B_{(L+1)R}$, let $z_r(x)$ be a nearest point of $(r/(100d))\mathbb{Z}^d$. The annulus determining $E_{z_r(x),r}$ is contained in $A_{r/4,8r}(x)$. The event remains measurable from the latter field modulo constants, even though its normalizing average is centered at $z_r(x)$: the difference between the two spherical averages is measurable there.

For a prescribed power $s > 0$, choose $A > (2d+s)\log(64)/\rho$. The shell input (E4), with success fraction $1/2$, gives a threshold p such that the probability of no successful scale at a fixed grid point is at most $e^{-A|\mathcal{T}|}$. This threshold depends on A and the fixed annular geometry, but not on L or the number of grid points. If $|\mathcal{T}| \geq \rho \log_{64} \varepsilon^{-1} - C_0$, the union bound is at most

$$C_L e^{AC_0} \varepsilon^{A\rho/\log(64)-2d} = O_{L,C_0}(\varepsilon^s).$$

This proves the stated independence of the choice of p. The rounding errors are much less than $r/2$, so the resulting balls cover the intervening points as well. □

Choose δ and $p_* > 0$ so small that

$$\mathbb{P}[F_r(z)] \geq 1-\delta, \quad \mathbb{P}[H_r(t)] < p_* \quad \Longrightarrow \quad \mathbb{P}[F_r(z) \setminus H_{z,r}(t)] \geq p,$$

where p is large enough for Lemma 5.2 with $\rho = 1/4$. These choices, and the regularity constants above, are independent of t. The independence of p from L is used here: the macro confinement radius in the frequency proof may be chosen later, after the witness probability and ratio thresholds are fixed.

Definition 5.3. For $u < U$ and $\beta \in (0,1)$, let $G_R(u,\beta)$ be the event that there are $x, y \in \overline{B}_{2R}$ such that

$$|x-y| \geq \beta R, \quad \beta s_R(0) \leq A_h(x,y) \leq \beta^{-1} s_R(0), \quad B_h(x,y) > u A_h(x,y).$$

Proposition 5.4 (Frequency lemma)**.** *There is $\theta > 0$, depending only on the two metrics and the regularity choices, with the following property. If*

$$0 < t < U, \qquad u > U - \theta(U-t), \qquad \mathbb{P}[G_R(u,\beta)] \geq \beta, \tag{6}$$

then, for every sufficiently small $\varepsilon > 0$, at least three quarters of the scales in $\mathcal{S} \cap [\varepsilon^2 R, \varepsilon R]$ satisfy

$$\mathbb{P}[H_r(t)] \geq p_*. \tag{7}$$

The threshold for ε is uniform in R when t, u, β are fixed.

Proof. Set $\theta = a/A_0$. Suppose at least one quarter of the scales violate (7). Lemma 5.2 gives a covering of B_{LR} by balls with $F_r(z) \setminus H_{z,r}(t)$ occurring. Choose L so large that confinement from Lemma 2.7 fails with probability less than $\beta/10$. The covering fails with probability tending to zero uniformly in R.

Consider any selected A_h-geodesic P with endpoints as in $G_R(u,\beta)$. Its parts before its last visit to $\overline{B}_{4\varepsilon R}(x)$ and after its first subsequent visit to $\overline{B}_{4\varepsilon R}(y)$ have total length $o_\varepsilon(s_R(0))$, uniformly for all endpoints, on a common modulus event. This follows from the geodesic property: a prefix ending near x has length equal to the endpoint distance, regardless of its Euclidean trace; likewise for a suffix.

Starting just after the last visit near x, take a covering ball $B_{r/2}(z)$ containing the current point and run P to its next exit from $B_{2r}(z)$. Both macro endpoints are outside that ball. The segment from the current point to that exit has length at most $A_0 s_r(z)$: enlarge it backwards to the previous visit to $\partial B_{2r}(z)$ and use the internal spherical diameter in (R1). The last crossing of $A_{r,2r}(z)$ before the exit has length at least $a s_r(z)$ and, by Lemma 5.1, admits a B_h replacement of ratio at most t.

Continue at the exit time. Stop at the first such step which passes the first subsequent visit near y. Its exit is within $8\varepsilon R$ of y, so the unprocessed suffix is still $o_\varepsilon(s_R(0))$. The procedure terminates after finitely many steps: there are finitely many possible scales and centers, and each processed step has length bounded below by a positive random minimum. The intervals are disjoint in time. Their replaced subintervals occupy at least a θ fraction of their total length. On all other portions use $B_h \le U A_h$. Hence

$$B_h(x,y) \le [U - \theta(U-t)] A_h(x,y) + o_\varepsilon(s_R(0)). \tag{8}$$

The error is uniform over the macro endpoints on the common modulus event. Since $A_h(x,y) \ge \beta s_R(0)$, (6) contradicts $G_R(u,\beta)$ for small enough ε. The total probability of the confinement, covering and modulus failures can be made less than β, contradicting the assumed probability of G_R. □

Corollary 5.5. *If U is the optimal essential upper comparison constant for the ordered pair, then for every $t < U$ the scales satisfying (7) occupy at least three quarters of every sufficiently small logarithmic window $[\varepsilon^2 R_0, \varepsilon R_0]$, for a suitable deterministic macro radius R_0. The macro radius can be enlarged in advance to accommodate any finite list of such applications.*

Proof. Choose u strictly between $U - \theta(U-t)$ and U. Optimality gives a positive-probability ratio witness, and continuity gives one with deterministic rational endpoints. Enlarge a deterministic ball to contain them. Since the normalized distance is positive and finite, some $\beta > 0$ gives $\mathbb{P}[G_{R_0}(u,\beta)] \ge \beta$. Enlarging R_0 only changes the permissible value of β. Apply Proposition 5.4. □

6 High-probability local traps from favorable arcs

Assume from now on that $c_* < C_*$. Prepare the regularity and frequency constants of Section 5 for both ordered pairs at this point. Use the smaller of their two positive p_* values throughout; all conclusions remain valid after decreasing p_*. These choices precede every parameter choice below. Apply Section 5 to the lower-ratio ordered pair $(A_h, B_h) = (\widetilde{D}_h, D_h)$, whose optimal upper constant is $1/c_*$. Choose

$$c_* < c_0 < c_1 < c_2 < c_3 < C_*. \tag{9}$$

First choose c_1, c_2, c_3; then choose c_0 close enough to c_* that

$$\frac{1}{c_0} > \frac{1}{c_*} - \theta_- \left(\frac{1}{c_*} - \frac{1}{c_1} \right), \tag{10}$$

where θ_- is the frequency constant for this ordered pair. The event $H_r^-(1/c_1)$ supplies a favorable $\widetilde{D}_h$-geodesic arc in $A_{r,2r}$ with

$$\operatorname{len}(\gamma; \widetilde{D}_h) < c_1 D_h(\gamma(0), \gamma(1)),$$

and with fixed normalized lower endpoint separation and upper length bounds. The superscript only specifies the ordered pair.

Lemma 6.1 (Independent fine-field amplification)**.** *Fix finitely many disjoint deterministic open boxes in $A_{1,3}$, with specified smaller compact subboxes, and $p < 1$. There is $\rho = 64^{-m}$ such that, whenever*

$$\mathbb{P}[H^-_{\rho r}(1/c_1)] \geq p_*, \tag{11}$$

with probability at least p every dilated compact subbox contains a simple rectifiable arc γ satisfying

$$\operatorname{len}(\gamma; \widetilde{D}_h) \leq c_2 D_h(\gamma(0), \gamma(1)), \qquad D_h(\gamma(0), \gamma(1)) \geq a_1 s_r(0), \qquad \operatorname{len}(\gamma; D_h) \leq A_1 s_r(0), \tag{12}$$

where $0 < a_1 < A_1 < \infty$ are deterministic. All choices are uniform in r satisfying (11).

Proof. Use the compact-kernel white-noise representation of the LGF from [1, Section 2.3]. For a smooth radial compactly supported kernel $\mathfrak{K}$ with $\int \mathfrak{K}^2 = 1$, the field, modulo constants, is represented by

$$h(x) = \int_0^\infty \int_{\mathbb{R}^d} t^{-(d+1)/2} \mathfrak{K}((x-y)/t)\, W(dy, dt).$$

At trial scale $s = \rho r$, let $F = h_{<Ls}$ be the stationary fine field and write $h = F + g$. The coarse field g is smooth modulo constants. Its class modulo constants is independent of F; normalizing h may change the chosen constant of g, which affects none of the oscillations below. If $R_{\mathfrak{K}}$ bounds the support radius of $\mathfrak{K}$, then $F|_{B_{5s}(w)}$ depends only on white noise with $t < Ls$ and spatial coordinate in $B_{(5+R_{\mathfrak{K}}L)s}(w)$. Disjoint such neighborhoods therefore give independent fine-field restrictions.

For every fixed-size trial annulus, the coarse oscillation tends to zero in probability as $L \to \infty$, uniformly in s and the center. For example, each derivative of order $k \geq 1$ has variance $O((Ls)^{-2k})$. After rescaling the trial ball to unit size and subtracting its value at the center, the expected squared Sobolev norm of any fixed sufficiently high integer order is $O(L^{-2})$. Sobolev embedding and Markov's inequality therefore give the claimed uniform oscillation bound. Choose $\delta > 0$ with

$$e^{4\xi\delta} c_1 < c_2, \qquad e^{4\xi\delta} < 2.$$

We now specify a local trial event, so that its independence does not require preserving geodesicity during a field shift. Write $\ell = \ell_-$ for the regular length constant of the ordered pair $(\widetilde{D}, D)$ and put

$$I = \overline{A}_{s,2s}(w), \quad O = A_{3s/4,3s}(w), \quad J_0 = A_{s/2,5s}(w), \qquad C_{\mathrm{col}} = 4 \max\{1, c_*^{-1}\}\ell.$$

For an input X, set $s_X = s^{\xi Q} e^{\xi X_s(w)}$. Define $\mathcal{W}_{w,s}(X)$ to mean that

$$D_X(I, \partial O; J_0) > e^{-\xi\delta} C_{\mathrm{col}} s_X \tag{13}$$

and that there is a rectifiable curve $P \subset I$, with endpoints u, v, satisfying

$$\begin{gathered} D_X(u, v; O) \geq e^{-\xi\delta} \frac{\ell}{2c_1} s_X, \qquad \operatorname{len}(P; \widetilde{D}_X) \leq e^{\xi\delta} \ell s_X, \\ \operatorname{len}(P; \widetilde{D}_X) \leq e^{2\xi\delta} c_1 D_X(u, v; O). \end{gathered} \tag{14}$$

Path lengths here can be computed from the internal J_0 coordinate. Thus all these conditions are determined by $X|_{J_0}$ and its spherical average in J_0.

On $\{\operatorname{osc}_{J_0} g \leq \delta\}$, the normalized fields $h - h_s(w)$ and $F - F_s(w)$ differ by $g - g_s(w)$, whose absolute value on J_0 is at most δ. Every normalized internal distance and curve length in these domains is therefore changed by a factor between $e^{-\xi\delta}$ and $e^{\xi\delta}$. A full-field $H^-_{w,s}(1/c_1)$ witness

has $\widetilde{D}_h$-length in $[\ell/2,\ell]s_h$ and $D_h(u,v;O) > c_1^{-1}\operatorname{len}(P;\widetilde{D}_h)$. Its regular D_h collar exceeds $C_{\mathrm{col}}s_h$. These facts transfer directly to (13)–(14) for $X = F$. Consequently

$$\mathbb{P}[\mathcal{W}_{w,s}(F)] \geq p_* - \mathbb{P}[\operatorname{osc}_{J_0} g > \delta].$$

The fine field is an LGF plus the possibly dependent random smooth function $-g$, precisely as allowed in the weak axioms. Locality thus makes $\mathcal{W}_{w,s}(F)$ a function of the restricted fine field. The event is unchanged by adding a constant to F. Trial events from the disjoint white-noise neighborhoods specified above are independent, even though their full-field counterparts need not be independent.

Let M be the number of boxes and choose N so large that $M(1-p_*/2)^N < (1-p)/3$. Next choose L so large that the oscillation failure probability at each of the MN trials is less than both $p_*/2$ and $(1-p)/(3MN)$. Finally choose $\rho = 64^{-m}$ sufficiently small to place all the trial regions and their disjoint fine-noise neighborhoods inside the specified compact subboxes. Each fine trial then succeeds with probability at least $p_*/2$. Independence and a union bound over the boxes, followed by the separate union bound on coarse oscillations, show that with probability at least $1 - 2(1-p)/3$ every box has a successful fine witness and all coarse oscillations are small. No independence from the latter events is used.

For a successful trial, transfer its curve back to h. The inequalities above give

$$\begin{aligned} D_h(u,v;O) &\geq e^{-2\xi\delta}\frac{\ell}{2c_1}s_h, \qquad & \operatorname{len}(P;\widetilde{D}_h) &\leq e^{2\xi\delta}\ell s_h, \\ \operatorname{len}(P;\widetilde{D}_h) &\leq e^{4\xi\delta}c_1 D_h(u,v;O), \qquad & D_h(I,\partial O;J_0) &> e^{-2\xi\delta}C_{\mathrm{col}}s_h. \end{aligned} \tag{15}$$

Since $D_h \leq c_*^{-1}\widetilde{D}_h$ for path lengths, P is a D_h competitor of cost at most $c_*^{-1}e^{2\xi\delta}\ell s_h$. This is less than the displayed collar bound, by $e^{4\xi\delta} < 2$. Any competitor starting at u which leaves O must pay that collar before its first exit; the part up to that exit lies in J_0. It follows that $D_h(u,v) = D_h(u,v;O)$. Thus (15) gives the required global endpoint ratio with the strict margin $e^{4\xi\delta}c_1 < c_2$.

The finitely many differences $h_s(w) - h_r(0)$ are centered Gaussian with variance $\log(1/\rho)+O(1)$, uniformly in r, since the trial centers divided by r range over a fixed finite set. Bound their absolute values by a large deterministic K with failure probability less than $(1-p)/3$. Since $s_h/s_r(0) = \rho^{\xi Q}e^{\xi(h_s(w)-h_r(0))}$, one may take

$$a_1 = \rho^{\xi Q}e^{-\xi K - 2\xi\delta}\frac{\ell}{2c_1}, \qquad A_1 = \rho^{\xi Q}e^{\xi K + 2\xi\delta}\frac{\ell}{c_*}.$$

For simplicity of the arcs, minimize $\widetilde{D}_h$-length within the compact image of each witness curve. A minimizer exists by reparameterization, compactness and lower semicontinuity, and is simple because its endpoints differ. Its endpoint ratio is preserved. Its D_h-length is at most c_*^{-1} times its $\widetilde{D}_h$-length, so the same A_1 still applies. This proves (12). □

Proposition 6.2 (Finite deterministic trap family)**.** *Given $p \in (0,1)$, there are an integer $m \geq 1$, a scale ratio $\rho = 64^{-m}$, a thin deterministic collector $V \Subset A_{1,3}$ containing ∂B_2, a finite family $f_1,\dots,f_J \in C_c^\infty(A_{1,3})$, all nonnegative and positive exactly on V, and constants*

$$a,b,c,g,H > 0, \quad c' < C_*, \quad \max_j \|f_j\|_\infty \leq M_f < \infty,$$

with the following properties. Write $f_{z,r,j}(x) = f_j((x-z)/r)$ and $V_{z,r} = z + rV$, and define

$$\mathcal{R} = \left\{ r \in \mathcal{S} : \mathbb{P}[H^-_{\rho r}(1/c_1)] \geq p_* \right\}. \tag{16}$$

At each $r \in \mathcal{R}$ there are events $E_{z,r,j}$, for every $z \in \mathbb{R}^d$, measurable modulo constants from the field on $A_{r/2,5r}(z)$, such that

$$\mathbb{P}\left[\bigcup_{j=1}^{J} E_{z,r,j}\right] \geq p.$$

On $E_{z,r,j}$ the following hold, with $s = s_r(z)$:

(T1) *Every path from $\overline{V}_{z,r}$ to either boundary of $A_{r,3r}(z)$ has original D_h-length at least as.*

(T2) *Every two points of $\overline{V}_{z,r}$ have internal $D_{h-f_{z,r,j}}$ distance in $A_{r,3r}(z)$ at most $(2a-g)s$, where $g = a$. In particular, this holds for every two points of $\partial B_{2r}(z)$.*

(T3) *Consider an internal $D_{h-f_{z,r,j}}$-geodesic in $B_{4r}(z)$. Suppose it meets $\overline{V}_{z,r}$, and let u, v be its first and last points in this closed set. If $D_h(u,v) \geq bs$, then between those times it contains a subsegment of length at least cs, whose endpoints are joined by a path in $B_{4r}(z)$ of $\widetilde{D}_{h-f_{z,r,j}}$-length at most c' times that length.*

(T4) *Let $\mathcal{Z}$ be a finite collection of labeled sites containing (z,j), with pairwise disjoint radius-$4r$ balls, and put $F = \sum_{(w,i)\in\mathcal{Z}} f_{w,r,i}$. For every D_{h-F}-geodesic, at most one excursion into $B_{4r}(z)$ meets $\overline{V}_{z,r}$. Its total original D_h-length spent in $V_{z,r}$ is at most Hs, with $H = ae^{\xi M_f}$.*

The constants satisfy

$$2b < g/2. \tag{17}$$

For collector endpoints, the threshold test in (T3) is unchanged if the original endpoint distance is replaced by its internal $B_{4r}(z)$ distance.

Proof. All choices are made in normalized units. We give their order explicitly. Begin with $V_0 = A_{7/4,9/4}$. By uniform tightness and nondegeneracy, choose $a > 0$ so that, with probability as close to one as prescribed, the U-coordinate distances from $\overline{V}_0$ to $\partial A_{1,3}$ are greater than $4a$. This choice precedes the collector, graph, and discount depth. Set

$$g = a, \qquad 0 < b < a/4.$$

The no-reentry argument below uses only the collar and the all-collector connection bound, so it is independent of depth. Choose $\varepsilon_0 < e^{-\xi} b/200$ with $3\varepsilon_0 < a/4$.

Uniform local upper moduli allow a sufficiently fine deterministic mesh of ∂B_2 and a sufficiently thin deterministic collector $V \Subset V_0$ so that, except on another event of arbitrarily small probability, each enlarged vertex star has U-coordinate D-diameter at most $b/16$, and

$$\max_{x\in\overline{V}} \min_{v \text{ a mesh vertex}} D(x, v; U_1) \leq \varepsilon_0/2. \tag{18}$$

Choose small central boxes in the edge tubes of a spanning tree of the mesh. Apply Lemma 6.1 to these fixed boxes, with a sufficiently high success probability. This fixes one $\rho = 64^{-m}$ and gives favorable witnesses in all the boxes at every scale in (16), with common constants a_1, A_1. Keep this ρ for the remainder of the proof and the subsequent counting argument. If necessary enlarge A_1 after loop erasure using $D \leq c_*^{-1}\widetilde{D}$. There are no further restrictions on how small a_1 or how large A_1 may be.

Take a compact set $\mathcal{C}$ of the four normalized internal metric coordinates on U, U_1, of uniformly high probability as in Lemma 2.2. Inside $\mathcal{C}$, impose the following closed conditions: the buffered

collar bounds $\geq 4a$; the star diameter bounds $\leq b/16$; (18); and, in every central compact box, existence of a path on $[0,1]$ with D^U-Lipschitz constant at most A_1, endpoint D^U-distance at least a_1, and

$$\operatorname{len}(\gamma; \widetilde{D}^U) \leq c_2 D^U(\gamma(0), \gamma(1)).$$

Call the resulting set $\mathcal{G}$. Its probability is at least p after allocating the preceding exceptional probabilities appropriately. It is compact. Indeed, the path fibers in the last conditions are compact by the common inverse modulus, and their constraints are closed: length is a supremum of continuous finite-partition sums. Projecting these compact fibers preserves closedness. All other constraints are continuous inequalities on fixed compact sets.

Every configuration of $\mathcal{G}$, including those occurring only as limits, has the paths needed for the construction. The U_1 access inequalities produce paths in $\overline{U_1} \Subset U$ by the final assertion of Lemma 2.2. A shortest $\widetilde{D}^U$ path in the compact image of each favorable witness is simple and preserves its endpoint ratio. Its D^U-length is at most $c_*^{-1} c_2 A_1$, since the original witness has endpoint D^U-distance at most A_1. Apply Lemma 4.1 with this finite arc-length bound in place of A_1; the bound defining the compact witness fibers is kept unchanged. The local length structure on U gives all routing paths, and the graph construction leaves a ratio margin from c_2 to c_3.

The following localization justifies the use of the local configuration throughout. A weighted path between collector points that exits $A_{1,3}$ pays at least $4a$ after its last collector visit before that exit: on this portion the discount is zero. In contrast, the access-and-graph competitor can be made to cost less than $3\varepsilon_0 < a/4$. Comparison shortcuts concatenate the marked arcs with the connection paths constructed in the collapse proof. These are concrete paths in the same confining annulus, and their weighted comparison lengths are controlled by Lemma 2.1. Under a perturbation of the metric configuration, these fixed compact competitors are recovered locally as in Lemma 4.2. Thus all paths used for collapse and shortcut transfer are inspected inside U. Original distances less than b between collector points also have their competitors inside $A_{1,3}$, up to arbitrarily small errors. This proves the asserted equivalence of threshold tests.

Choose $c_3 < c' < C_*$. For each configuration of $\mathcal{G}$, Lemma 4.2 gives one smooth discount and the shortcut assertion with endpoint threshold $b/2$, strict ratio below c', and a positive length margin. It also gives connection costs below $a/2$ between all collector points. By fixed-weight stability, these properties with final threshold b, a smaller positive length constant, and final ratio c' hold on a relative neighborhood in $\mathcal{C}$. The buffered collar bound persists there with constant a. A finite subcover of $\mathcal{G}$ gives deterministic $f_1, \ldots, f_J$. Let c be the minimum of the finitely many positive length constants and M_f the maximum amplitude. Define $E_{z,r,j}$ by membership in $\mathcal{C}$ and the corresponding relative neighborhood, with the final collar bound. These are Borel events determined by the annular field modulo constants. Their union contains $\mathcal{G}$, so has probability at least p. This proves (T1)–(T3).

It remains to check (T4). Consider a geodesic for simultaneous disjoint discounts. The segment between its first and last visits to $\overline{V}_{z,r}$ has weighted length at most as, by (T2). The competing connection stays in $A_{r,3r}(z)$, so the other discounts do not alter it. If two different $B_{4r}(z)$ excursions met the closed collector, the portion between them would pay an unaffected collar on departure and another on return. More precisely, use the last collector visit before an exit and the first one after the subsequent return, and restrict each portion up to or from $\partial B_{3r}(z)$. Both portions lie inside $B_{4r}(z)$ and outside $\overline{V}_{z,r}$ except at their collector endpoints. Thus every discount is zero along their interiors, and each has original length at least as by (T1). Their weighted total is therefore at least $2as$, contradicting the preceding as bound. This proves the single-excursion assertion.

On the whole segment from the first to the last collector visit, the weight in $V_{z,r}$ is at least $e^{-\xi M_f}$. Lemma 2.1 consequently bounds the total original length spent there by $ae^{\xi M_f} s$. This

proves (T4). Only H depends on the depth; the previously fixed inequality (17) is retained. □

Corollary 6.3 (Dense trap scales below a regular lower-ratio scale)**.** *For sufficiently small ε, uniformly over all R satisfying*

$$\mathbb{P}[H_R^-(1/c_0)] \geq p_*, \tag{19}$$

at least two thirds of the scales in $\mathcal{S} \cap [\varepsilon^2 R, \varepsilon R]$ belong to $\mathcal{R}$.

Proof. The regular event in (19) implies $G_R(1/c_0, \beta)$ for a fixed $\beta > 0$ with probability at least β. Equation (10) permits Proposition 5.4 with target $1/c_1$. This gives three-quarters density of the corresponding good arc scales. Replacing a scale s by $r = s/\rho$ shifts its logarithmic index by the fixed integer m. At most $2m + O(1)$ indices are lost at the ends of the window. For sufficiently small ε the remaining density is at least two thirds. □

7 Gaussian change of measure and a common covering event

Let $\mathcal{H}$ be the Cameron–Martin space of the LGF modulo constants. For smooth compactly supported functions its norm is

$$\|f\|_{\mathcal{H}}^2 = a_d \int_{\mathbb{R}^d} |\zeta|^d |\widehat{f}(\zeta)|^2 \, d\zeta, \qquad a_d > 0. \tag{20}$$

Translation and dilation preserve this critical norm. For the fixed finite smooth family in Proposition 6.2, there is therefore $M_0 < \infty$ with $\|f_{z,r,j}\|_{\mathcal{H}} \leq M_0$ for every z, r, j. In particular, if F_Z is a sum of at most k such functions, then

$$\|F_Z\|_{\mathcal{H}}^2 \leq M_0^2 k^2. \tag{21}$$

This bound uses only the triangle inequality, in every dimension.

Lemma 7.1 (A change-of-measure inequality)**.** *For a measurable event A, a deterministic $F \in \mathcal{H}$, and $p_H > 1$,*

$$\mathbb{P}[A(h)] \leq \exp\left\{ \frac{p_H - 1}{2} \|F\|_{\mathcal{H}}^2 \right\} \mathbb{P}[A(h + F)]^{1 - 1/p_H}. \tag{22}$$

Proof. Write μ for the Gaussian law modulo constants and μ_F for the law of $h + F$. Its density is

$$L_F = \frac{d\mu_F}{d\mu} = \exp\{W(F) - \tfrac{1}{2}\|F\|_{\mathcal{H}}^2\},$$

where $W(F)$ is centered Gaussian with variance $\|F\|_{\mathcal{H}}^2$. Hölder's inequality under μ_F gives

$$\mu(A) \leq \mu_F(A)^{1 - 1/p_H} \left(\mathbb{E}_{\mu_F} L_F^{-p_H} \right)^{1/p_H}.$$

The last expectation equals $\exp\{p_H(p_H - 1)\|F\|_{\mathcal{H}}^2/2\}$ by the Gaussian exponential moment formula. This proves the assertion, also for completed Gaussian sigma-algebras because the two laws are equivalent. □

Lemma 7.2 (A common geometric covering event)**.** *Choose p in Proposition 6.2 sufficiently close to one. For every macro scale R satisfying (19), with probability tending to one polynomially as $\varepsilon \to 0$, every point of B_{5R} lies in a ball $B_{r/2}(z)$ with*

$$r \in \mathcal{R} \cap [\varepsilon^2 R, \varepsilon R], \qquad z \in \frac{r}{100d} \mathbb{Z}^d, \qquad E_{z,r,j} \text{ for at least one } j. \tag{23}$$

The rate is uniform in R, and this event is independent of the eventual endpoint pair and of the number of perturbation sites.

Proof. Corollary 6.3 gives a fixed positive density of admissible scales. Apply Lemma 5.2 to the union of the local events at each such scale. The probability threshold p is independent of the fixed logarithmic scale shift m; that shift changes only how small ε must be. □

8 A perturbation counting estimate

The comparison-by-counting architecture follows [4, Section 4]. We give the needed version explicitly, since the finite graph traps and the finite family of perturbations require explicit bookkeeping. Fix $0 < \eta \le 1$ and deterministic $x, y \in B_{3R}(0)$ with $|x - y| \ge \eta R$. Throughout this section take $0 < \varepsilon < 1/3$ and restrict every perturbation scale index r, and all outer counting sums over scales, to

$$r \in \mathcal{R} \cap [\varepsilon^2 R, \varepsilon R].$$

Put

$$S = R^{\xi Q} e^{\xi h_{8R}(0)}, \qquad D = D_h(x, y), \qquad \widetilde{D} = \widetilde{D}_h(x, y). \tag{24}$$

All bumps used below have support in B_{6R}, so S is unchanged by adding any of them. Their individual normalizers $s_r(z)$ are unchanged as well: each bump vanishes on its own radius-r sphere, and distinct chosen centers are separated by at least $10r$.

Lemma 8.1 (A global scale bin event)**.** *There is a deterministic $B > 0$ such that, uniformly in R, with probability tending to one polynomially as $\varepsilon \to 0$,*

$$\varepsilon^B \le s_r(z)/S \le \varepsilon^{-B} \tag{25}$$

for every $r \in \mathcal{S} \cap [\varepsilon^2 R, \varepsilon R]$ and every $z \in (r/(100d))\mathbb{Z}^d \cap B_{5R}$.

Proof. The centered Gaussian variable $h_r(z) - h_{8R}(0)$ has variance at most $\log(R/r) + O(1) \le 2\log\varepsilon^{-1} + O(1)$, uniformly over these centers. This follows directly by integrating the logarithmic covariance against the two spherical measures; the radius-r sphere stays a fixed positive multiple of R away from ∂B_{8R}. There are $O(\varepsilon^{-2d}\log\varepsilon^{-1})$ variables. A Gaussian tail bound and a union bound show that their absolute values are at most $B_0 \log\varepsilon^{-1}$ with polynomially high probability, for a sufficiently large B_0. Equation (25) follows from the definition of $s_r(z)$, taking $B > 2\xi Q + \xi B_0 + 1$. □

At a fixed scale, partition $(r/(100d))\mathbb{Z}^d$ into finitely many colors, each with spacing $10r$. All radius-$4r$ balls with centers of one color are disjoint. There are $(1000d)^d$ colors, independent of scale.

For such a scale r, a color, and a dyadic q in the deterministic range $[\varepsilon^B/2, \varepsilon^{-B}]$, a labeled set is

$$Z = \{(z_1, j_1), \dots, (z_n, j_n)\}, \qquad F_Z = \sum_{i=1}^{n} f_{z_i, r, j_i},$$

with distinct centers of that color in B_{5R}, and with $x, y \notin \overline{B}_{4r}(z_i)$. Let P_h and $P_{h - F_Z}$ be the selected geodesics between x and y. Define $\mathcal{F}_Z(h)$ by the following conditions:

(C1) $\widetilde{D} \ge C_* D - qS$.

(C2) For every $(z_i, j_i) \in Z$, E_{z_i, r, j_i} occurs and $s_r(z_i) \in [qS, 2qS]$.

(C3) P_h visits $B_r(z_i)$ for every i.

(C4) For every i, P_{h-F_Z} has a $B_{4r}(z_i)$ excursion visiting $V_{z_i,r}$ whose first and last $\overline{V}_{z_i,r}$ points have original internal D_h-distance in $B_{4r}(z_i)$ at least $bs_r(z_i)$.

Lemma 8.2 (Exact identities for separated perturbations)**.** *For a fixed labeled set Z, put $f_i = f_{z_i,r,j_i}$. Almost surely, for every $i \in Z$,*

$$(h + F_Z)_r(z_i) = h_r(z_i), \qquad (h + F_Z)_{8R}(0) = h_{8R}(0),$$

and, for either metric and every open $W \subset B_{5r}(z_i)$,

$$D_{h+F_Z}(\cdot,\cdot;W) = D_{h+f_i}(\cdot,\cdot;W).$$

Consequently

$$E_{z_i,r,j_i}(h + F_Z) = E_{z_i,r,j_i}(h + f_i), \qquad P_{(h+F_Z)-F_Z} = P_h.$$

All conditions (C1)–(C4) are invariant under adding a constant to the field. These conclusions therefore also apply when a shifted field is returned to the representative with unit-sphere average zero.

Proof. Every f_i vanishes on its own radius-r sphere. For $j \neq i$, separation of the centers by at least $10r$ and support of f_j in $B_{3r}(z_j)$ imply that f_j is zero throughout $B_{5r}(z_i)$. Hence the first spherical average is unchanged. Also $|z_i| < 5R$ and $3r < R$, so every support lies in B_{6R} and misses ∂B_{8R}. This proves the second average identity.

On W, the continuous function $F_Z - f_i$ is zero. Simultaneous Weyl scaling and Lemma 2.1 therefore make all internal path lengths there identical for $h + F_Z$ and $h + f_i$. This proves the internal-metric identity for both metrics. The local event E_{z_i,r,j_i} is a membership test on the corresponding normalized annular metric coordinates. Those coordinates and their spherical normalizer are identical, which gives the event identity. Subtracting the full deterministic perturbation from $h + F_Z$ gives h itself, proving the path identity.

Adding a constant a multiplies both metrics, every internal distance and every normalizer s_r or S by $e^{\xi a}$. The selected geodesic is unchanged by this multiplication. Each condition (C1)–(C4) is therefore unchanged. In particular, if $h_1(0) = 0$, the normalized shifted field is $h + F_Z - (F_Z)_1(0)$. Discounting it by F_Z gives $h - (F_Z)_1(0)$, whose selected geodesic is again P_h. The Cameron–Martin formula can thus be applied to the class of F_Z in the Gaussian field modulo constants, without introducing a random normalization factor into any counting event. □

The corresponding internal-metric identity also holds for negative shifts. In particular, take a $B_{4r}(z_i)$ excursion of P_{h-F_Z} which visits the collector, and restrict it to the segment between its first and last closed collector visits. This segment is compactly contained in $B_{4r}(z_i)$. Global minimality implies that it minimizes length internally there, and its internal path lengths are those for $h - f_i$, since all other bumps vanish in the ball. Thus, whenever its original endpoint distance meets the stated threshold, (T3) applies to this segment. No geodesic with endpoints on the boundary of the open ball is needed. Applied with base input $h + F_Z$, the same observation gives precisely the local use of the fixed path P_h in the shifted count.

All deterministic bumps and their finite sums form a countable family, so their almost-sure identities can be imposed on one common full-probability event. We use completed Gaussian sigma-algebras. All path spaces are Polish with the local uniform topology. The compact-geodesic selection described after Lemma 2.7, and existential quantification over paths and times, give universally measurable events. This suffices for the change-of-measure arguments.

Lemma 8.3 (Pruning retains a positive fraction)**.** *Suppose a set of n labeled sites satisfies (C2)–(C3). Then it has a subset Z' of at least κn sites satisfying (C4), where $\kappa > 0$ is deterministic. Consequently it satisfies all of $\mathcal{F}_{Z'}(h)$ whenever (C1) holds.*

Proof. Write $s_i = s_r(z_i)$ and

$$\Delta(Z) = D_h(x, y) - D_{h-F_Z}(x, y).$$

For each site choose the component of $P_h \cap B_{2r}(z_i)$ containing a visit to $B_r(z_i)$. Its endpoints lie on $\partial B_{2r}(z_i)$ and its original length is at least $2as_i$. Replace it by the discounted connection in (T2). These components are disjoint in time because their balls are disjoint. Elsewhere the discounts only shorten the original path. Therefore

$$\Delta(Z) \geq g \sum_{i \in Z} s_i \geq gnqS. \tag{26}$$

If a site fails (C4) for the current discounted geodesic, undo that site. By (T4), at most one $B_{4r}(z_i)$ excursion visits $V_{z_i,r}$. In that excursion, replace the portion between the first and last closed collector visits by an original internal path of cost less than bs_i; failure of (C4) gives precisely this distance bound. The replacement avoids every other bump, and the rest of the path is unchanged. If there is no such excursion, the discount changes no path length. Thus undoing the site increases the distance by at most $bs_i \leq 2bqS$.

Continue until all surviving sites satisfy (C4). By (17), the resulting set Z' satisfies

$$\Delta(Z') \geq (g - 2b)nqS \geq (g/2)nqS.$$

On the other hand, restore the original weight along the final discounted geodesic. The occupation bound (T4) gives

$$\Delta(Z') \leq H \sum_{i \in Z'} s_i \leq 2H|Z'|qS.$$

Hence $|Z'| \geq \kappa n$ with $\kappa = g/(4H) = e^{-\xi M_f}/4 > 0$. The surviving geometric and original-geodesic conditions do not change during pruning. □

Lemma 8.4 (Many unshifted favorable sets)**.** *On the common covering event of Lemma 7.2 and bin event of Lemma 8.1, if*

$$\widetilde{D} \geq C_* D - \varepsilon^{B+1} S, \tag{27}$$

then, for all sufficiently small ε, simultaneously for every integer $\lceil 1/\kappa \rceil \leq k \leq \lfloor \varepsilon^{-1/2} \rfloor$,

$$\sum_{r,\,\mathrm{color},\,q} \sum_{1 \leq |Z| \leq k} \mathbf{1}_{\mathcal{F}_Z(h)} \geq (\varepsilon^{-1/2}/k)^{\lceil \kappa k \rceil}. \tag{28}$$

The outer sum has $O(\log^2 \varepsilon^{-1})$ terms, up to the fixed number of colors. The rate is uniform in R, x, y with the stated separation.

Proof. Take the portion of P_h which first crosses Euclidean radii from $\eta R/4$ to $\eta R/2$ around x. At first hitting times of levels separated by $20\varepsilon R$, choose covering balls from (23). The resulting centers are mutually separated by more than $8\varepsilon R$, lie in a fixed bounded macro ball contained in B_{5R}, and are uniformly away from both endpoints. There are at least $c_\eta \varepsilon^{-1}$ of them. Each is hit by P_h in $B_r(z)$ and has a valid type.

Pigeonhole the scale, color, and dyadic bin for $s_r(z)/S$. This leaves at least $N \geq \varepsilon^{-1/2}$ sites with common r, q, for sufficiently small ε. Their radius-$4r$ balls are disjoint. Since $q \geq \varepsilon^B/2$, (27) implies (C1). The bound on ε needed for this pigeonhole estimate is independent of k. Since $k \leq \varepsilon^{-1/2} \leq N$, every k-subset of these N sites can be pruned by Lemma 8.3 to a favorable subset of size between $m = \lceil \kappa k \rceil$ and k. A fixed output of size at least m belongs to at most $\binom{N-m}{k-m}$ input sets. Consequently the number of distinct outputs is at least

$$\frac{\binom{N}{k}}{\binom{N-m}{k-m}} = \frac{(N)_m}{(k)_m} \geq (N/k)^m.$$

This proves (28). □

Lemma 8.5 (Few shifted favorable sets)**.** *For fixed r, q and color, and for every field realization outside a common null set,*

$$\sum_{1 \leq |Z| \leq k} \mathbf{1}_{\mathcal{F}_Z(h+F_Z)} \leq K_0^k \tag{29}$$

for a deterministic $K_0 < \infty$ independent of r, q, R, x, y, k.

Proof. Call a label (z, j) very good if $E_{z,r,j}(h + f_{z,r,j})$ occurs, $s_r(z) \in [qS, 2qS]$, and P_h has a large excursion measured by $D_{h+f_{z,r,j}}$ as in (C4). If $\mathcal{F}_Z(h + F_Z)$ occurs, every label of Z is very good. By Lemma 8.2, its local event and original internal excursion distance reduce to those of $h + f_{z,r,j}$, its normalizers are unchanged, and its discounted selected geodesic is exactly P_h.

If no shifted event occurs, there is nothing to prove. Otherwise fix one such set Z_0 with $|Z_0| \leq k$. At every very good center not in Z_0, apply (T3) to the base geodesic P_h, viewed locally as the discounted geodesic for $h + f_{z,r,j}$. It has a subsegment of D_h-length at least cqS and a local $\widetilde{D}_h$ shortcut of ratio at most c'. Choose one type per center. These subsegments are disjoint in time; their balls are disjoint from all supports of F_{Z_0}. The shortcuts therefore remain valid for the field $h + F_{Z_0}$.

The occupation bound, now applied to $E_{z,r,j}(h + f_{z,r,j})$, gives

$$\operatorname{len}(P_h; D_{h+F_{Z_0}}) \leq D_h(x, y) + H \sum_{i \in Z_0} s_i \leq D_{h+F_{Z_0}}(x, y) + 2Hkq S.$$

Use the shortcuts at N distinct very good centers outside Z_0 and the inequality $\widetilde{D}_{h+F_{Z_0}} \leq C_* D_{h+F_{Z_0}}$ along the remaining portions of P_h. The near-extreme inequality for $\mathcal{F}_{Z_0}(h + F_{Z_0})$ yields

$$(C_* - c')cqS\, N \leq 2C_* Hkq S + qS.$$

Thus the total number of very good centers is at most a constant times k. Each has at most J labels. Every shifted favorable set is a subset of these labels, so the number of such sets of size at most k is at most $2^{K_1 Jk}$. This is (29). □

Proposition 8.6 (Superpolynomial truncated near-extreme bound)**.** *Let R satisfy (19), and let $\mathcal{A}_R^\varepsilon$ be the intersection of the common geometric covering and bin events. For every $N > 0$,*

$$\mathbb{P}\left[\mathcal{A}_R^\varepsilon,\ \widetilde{D}_h(x, y) \geq C_* D_h(x, y) - \varepsilon^{B+1} S\right] = O_N(\varepsilon^N). \tag{30}$$

The bound is uniform over the lower-good macro scales R and deterministic endpoint pairs with the prescribed positive relative Euclidean separation. The common event does not depend on these endpoints or on N.

Proof. Fix a scale, color and dyadic bin. At most $C_d\varepsilon^{-2d}$ possible centers occur, since $r \geq \varepsilon^2 R$. With J labels per center, the number M of possible labeled sets of size at most k is at most

$$M \leq (1 + JC_d\varepsilon^{-2d})^k \leq (C\varepsilon^{-2d})^k.$$

One can encode each set by an ordered list padded to length k with a dummy symbol; this proves the first inequality without an assumption relating k to the number of centers.

Apply Lemma 7.1 separately to each $\mathcal{F}_Z$ with its shift F_Z. Then apply Hölder's inequality to the finite sum over these sets. Equations (21) and (29) yield

$$\begin{aligned}\sum_Z \mathbb{P}[\mathcal{F}_Z(h)] &\leq e^{A_p k^2} M^{1/p_H} \left(\sum_Z \mathbb{P}[\mathcal{F}_Z(h + F_Z)]\right)^{1-1/p_H} \\ &\leq \exp\{A_p k^2 + C_p k\}\varepsilon^{-2dk/p_H}, \qquad A_p = \frac{p_H - 1}{2} M_0^2,\end{aligned}$$

with $C_p < \infty$ independent of R, x, y, k, ε. Different sets have different shifts: the first inequality is applied to each separately, whereas the shifted count is pointwise under the common base field h.

Sum over the $O(\log^2 \varepsilon^{-1})$ scale, color and bin choices, and use Lemma 8.4. For every integer $\lceil 1/\kappa \rceil \leq k \leq \lfloor \varepsilon^{-1/2} \rfloor$,

$$\begin{aligned}&\mathbb{P}[\mathcal{A}_R^\varepsilon,\ (27)] \\ &\quad \leq C_0 \log^2 \varepsilon^{-1} \exp\{A_p k^2 + C_p k\}\varepsilon^{-2dk/p_H} (k\sqrt{\varepsilon})^{\lceil \kappa k \rceil}.\end{aligned} \tag{31}$$

The upper bound on ε in the counting lemmas is independent of k.

Choose $p_H > 8d/\kappa$, after the trap family has been fixed, and set

$$b_0 = \frac{\kappa}{2} - \frac{2d}{p_H} > \frac{\kappa}{4} > 0.$$

For a prescribed N, fix an integer $k \geq \lceil 1/\kappa \rceil$ with $b_0 k > N + 1$. For sufficiently small ε this fixed k lies in the admissible range. The right side of (31) is then at most $C_{k,p_H} \log^2 \varepsilon^{-1} \varepsilon^{b_0 k}$, which is $O_N(\varepsilon^N)$. The quadratic Cameron–Martin cost is a finite constant because k is fixed before ε tends to zero. No Gaussian probe truncation or orthogonality is used. □

9 Closing the comparison gap

Apply Section 5 also to the ordered pair $(D_h, \widetilde{D}_h)$, and write its regular event as $H_R^+(t)$. Its constants ℓ_+, β_+ are fixed independently of $t < C_*$. We claim that there is a deterministic $t < C_*$ such that

$$\mathbb{P}[H_R^+(t)] < p_* \quad \text{for every } R \text{ satisfying (19).} \tag{32}$$

Both ordered pairs use the common p_* fixed in Section 6.

Choose $L < \infty$ so that, uniformly in R,

$$\mathbb{P}[L^{-1} \leq s_R(0)/S \leq L] > 1 - p_*/10. \tag{33}$$

The logarithm of this ratio is a centered Gaussian of fixed variance. Choose $0 < \alpha < \xi(Q - \sqrt{2d})$ and ν with $\nu\alpha > B + 2$. On the common upper-modulus event, approximating an endpoint in B_{2R} by a point of an $\varepsilon^\nu R$ grid changes its D_h-distance by at most

$$L\, O(\varepsilon^{\nu\alpha}) S.$$

The change in $\widetilde{D}_h$-distance is at most C_* times this. The upper-modulus event has probability tending to one uniformly in R. This assertion also holds with the mesh multiplied by a fixed constant accounting for Euclidean rounding.

Take ε sufficiently small and then put

$$t = C_* - \frac{\varepsilon^{B+1}}{4\ell_+ L}. \tag{34}$$

On $H_R^+(t)$, the witness endpoints satisfy $|x-y| \geq \beta_+ R$ and $D_h(x,y) \leq \ell_+ s_R(0)$. On (33), their additive comparison gap is at most $\varepsilon^{B+1}S/4$. The nearest grid endpoints x', y' are in B_{3R} and separated by at least $\beta_+ R/2$. The triangle inequality and $\widetilde{D}_h \leq C_* D_h$ give

$$C_* D_h(x',y') - \widetilde{D}_h(x',y') \leq C_* D_h(x,y) - \widetilde{D}_h(x,y) + 2C_*\big(D_h(x,x') + D_h(y,y')\big).$$

For sufficiently small ε, the right side is at most $\varepsilon^{B+1}S$. Thus the regular high-ratio event produces a grid pair to which Proposition 8.6 applies.

There are $O(\varepsilon^{-2\nu d})$ grid pairs. First choose $N > 2\nu d + 1$ in Proposition 8.6, including its fixed choice of k, and then decrease ε as needed. By (30), the probability of any such pair on $\mathcal{A}_R^\varepsilon$ is $O_N(\varepsilon^{N-2\nu d})$, which tends to zero uniformly over the lower-ratio macro scales R. The failures of the covering, bin, upper-modulus and macro-normalizer events are common to all pairs and are added *once*, not once per pair. Their total probability, together with the grid-pair bound, is less than p_* for a sufficiently small fixed ε. This proves (32).

Keep the chosen ε and hence t fixed. Apply Corollary 5.5 to the lower ordered pair with threshold $1/c_0$ and to the upper ordered pair with threshold t. Choose a common enlarged deterministic reference radius R_0 for these two applications. For a new, independent parameter $\delta \downarrow 0$, put

$$\mathcal{S}_\delta = \mathcal{S} \cap [\delta^2 R_0, \delta R_0].$$

For every sufficiently small δ, the lower-good scales satisfying (19) and the upper-good scales satisfying $\mathbb{P}[H_R^+(t)] \geq p_*$ each occupy at least three quarters of $\mathcal{S}_\delta$. This finite set is nonempty for small δ, so the two sets intersect. At a scale in their intersection, (32), which was proved uniformly in R using the already fixed ε, contradicts the upper-good inequality. There is no requirement to choose that ε again after t or R_0 has been fixed.

Hence $c_* < C_*$ is untenable. This proves Theorem 1.4, namely $\widetilde{D}_h = c_* D_h$ simultaneously for all points. Simultaneous Weyl scaling extends it to every continuous perturbation with the same constant.

10 Exact covariance and convergence

This section records the consequences of the auxiliary weak uniqueness statement proved in the preceding argument.

Lemma 10.1 (Strong candidates belong to the weak class)**.** *Every assignment satisfying the four strong axioms belongs to the auxiliary weak class, with normalizers $r^{\xi Q}$.*

Proof. First let h be an LGF. Exact covariance and Weyl scaling give

$$r^{-\xi Q} e^{-\xi h_r(0)} D_h(r\cdot, r\cdot) = D_{h(r\cdot) - h_r(0)}.$$

The field on the right has the law of an LGF normalized by its unit-sphere average. Thus these metric laws are identical for every $r > 0$, and the same is true jointly for internal metrics on any fixed list of domains.

Now write the input as $h + f$ with f a possibly dependent random continuous function. In normalized coordinates the weight is $e^{\xi(f(r\cdot)-f_r(0))}$. For $0 < r < R$, on each fixed compact set this weight and its reciprocal have a finite random common bound, using the supremum of $|f|$ on a larger compact set. Upper bounds and moduli for internal distances therefore follow from the preceding identical-law family. For a lower modulus, a path joining Euclidean-separated points in a compact set must first exit a small Euclidean ball about its initial point. Until that exit the weight has the stated positive lower bound, and the unweighted crossing cost has a uniform positive lower modulus. This applies regardless of where the path goes afterwards. Tight upper and lower moduli on a countable compact exhaustion give tightness and ensure that every limit is a finite metric with the Euclidean topology. When f is bounded, the global factor $e^{2\xi\|f\|_\infty}$ gives the same bounds uniformly for all $r > 0$. This is exactly the weak scale-tightness axiom. □

Corollary 10.2 (Exact dilation covariance)**.** *Every weak assignment satisfies, for each fixed $r > 0$, almost surely for all endpoints,*

$$D_h(rx, ry) = D_{h(r\cdot)+Q\log r}(x, y).$$

The identity holds after every continuous perturbation of h.

Proof. For fixed $r > 0$ define

$$(T_r D)_h(x, y) = r^{-\xi Q} D_{h(\cdot/r)}(rx, ry).$$

It satisfies the weak axioms. For scale tightness, if the normalizers of D are $\mathfrak{c}_s$, use $r^{-\xi Q}\mathfrak{c}_{rs}$ for $T_r D$; the normalized scale-s metric is precisely the normalized scale-rs metric of D evaluated at the transformed field. Locality, translation covariance and Weyl scaling follow by a change of coordinates.

Weak uniqueness gives a deterministic $a(r) > 0$ with $T_r D = a(r)D$, including after continuous perturbations. Since $T_s T_r = T_{sr}$, these constants satisfy $a(rs) = a(r)a(s)$. For $h_1(0) = 0$, the normalized scale-r metric

$$r^{-\xi Q} e^{-\xi h_r(0)} D_h(r\cdot, r\cdot)$$

has the law of $a(r)D_h$, because $h(r\cdot) - h_r(0)$ has the law of h. The sharp-normalizer input and nondegeneracy force deterministic constants $0 < a_- \le a(r) \le a_+ < \infty$, uniformly in r. An unbounded sequence of $a(r)$ would violate tightness of a positive point distance; a sequence tending to zero would produce the zero pseudometric as a limit. Applying the bounds to $a(r^n) = a(r)^n$ gives $a(r) = 1$. Replacing the input field by $h(r\cdot)$ gives the displayed identity. Weyl scaling extends it simultaneously to all continuous perturbations. □

Deduction of the four-axiom characterization. The subsequence theorem supplies a weak assignment. The preceding corollary upgrades it to exact dilation covariance; translation covariance is already a weak axiom. This proves existence in the strong class. By Lemma 10.1, two strong candidates are weak candidates, so auxiliary uniqueness proves their deterministic proportionality. This establishes the axiomatic assertion of Theorem 1.2. □

Lemma 10.3 (Full support of a normalized point distance)**.** *For $h_1(0) = 0$, the law of $D_h(0, e_1)$ gives positive probability to every nonempty open subinterval of $(0, \infty)$. It therefore has a unique median.*

Proof. Choose smooth compactly supported $0 \le f \le 1$, equal to one on a thin connected neighborhood U of $[0, e_1]$, with unit-sphere average $f_1(0) < 1$. Set $g = f - f_1(0)$ and $a = 1 - f_1(0) > 0$. For each fixed real t, the normalized laws of $h + tg$ and h are equivalent by the Cameron–Martin theorem applied to the compactly supported shift tf modulo constants.

For every realization, Weyl scaling and boundedness of g make $t \mapsto D_{h+tg}(0, e_1)$ continuous. Rectifiable connectivity of U gives

$$D_{h+tg}(0, e_1) \le e^{\xi a t} D_h(0, e_1; U) \longrightarrow 0 \qquad (t \to -\infty).$$

Choose a small ball $B_\delta(0) \Subset U$ excluding e_1. Every path to e_1 pays, before its first exit from this ball, at least $e^{\xi a t} D_h(0, \partial B_\delta(0))$, which tends to infinity as $t \to \infty$. The original distance in this bound is strictly positive. Thus every prescribed interval $(u, v) \Subset (0, \infty)$ is hit for some rational t. The union of the corresponding shifted events has probability one, so one shifted event has positive probability. Equivalence of laws proves the claim for the unshifted field. If two different numbers were medians, the interval between them would have probability zero, contradicting full support. □

Corollary 10.4 (Full LFPP convergence with a fixed normalization)**.** *For each convolution scheme in [1, Remark 1.1], normalize by a_ε, a median of $D_h^\varepsilon(0, e_1)$ for $h_1(0) = 0$. Then $\mathsf{a}_\varepsilon^{-1} D_h^\varepsilon$ converges in probability locally uniformly as $\varepsilon \to 0$ to the unique median-one weak, hence strong, assignment. With the same deterministic normalizers, convergence also holds for each LGF plus a bounded continuous function.*

Proof. By [1, Theorem 1.3], every sequence tending to zero has a deterministic subsequence converging in probability to a weak assignment. The closed-half-line inequalities in the portmanteau theorem show that one is a median of each limiting point distance. Weak uniqueness makes any two such assignments deterministic multiples. The preceding lemma gives a unique median, so that multiple must be one. If full convergence in probability failed, a sequence remaining a positive probability away from one fixed limit would have a further subsequence converging in probability to that same limit, a contradiction. The input theorem uses the same subsequence assignment for each LGF plus a bounded continuous function; simultaneous Weyl scaling identifies these assignments with the same scalar. □

A A direct Gaussian verification of the shell input

This appendix gives an independent verification of the high-probability part of the shell input. It uses Gaussian Hölder inequality and does not require a radial Markov property.

Lemma A.1 (Gaussian shell decoupling)**.** *Fix $0 < a < 1 < b$, $q \in (0, 1)$, and radii $r_{i+1} \le q r_i$. Let E_i be measurable with respect to $(h - h_{r_i}(0))|_{A_{ar_i, br_i}}$. For every $A > 0$ and $\theta \in (0, 1)$ there is $p < 1$, depending only on these parameters and d, such that*

$$\mathbb{P}[E_i] \ge p \quad \textit{for every } i \quad \Longrightarrow \quad \mathbb{P}\left[\sum_{i=1}^{n} \mathbf{1}_{E_i} < \theta n\right] \le e^{-An}.$$

Proof. For a compactly supported zero-mass distribution ν of finite logarithmic energy, write

$$\|\nu\|_-^2 = \mathrm{Var}(h, \nu) = \kappa_d \int |\zeta|^{-d} |\widehat{\nu}(\zeta)|^2 \, d\zeta.$$

This norm is unchanged by mass-preserving dilation. A linear observable of the normalized field on the ith annulus has the form (h, ν_i), where

$$\nu_i = \phi_i - \left(\int \phi_i \right) \sigma_{r_i}, \qquad \phi_i \in C_c^\infty(A_{ar_i, br_i}).$$

Here σ_r is uniform probability measure on ∂B_r. These measures have finite energy, and sphere averages are obtained by Gaussian L^2 limits. After rescaling, the supports lie in a fixed compact annulus bounded away from zero.

We first prove that, for two such unit-scale zero-mass distributions ν, μ, and sufficiently small ε,

$$|\operatorname{Cov}((h, \nu_\varepsilon), (h, \mu))| \le C\varepsilon \|\nu\|_- \|\mu\|_-. \tag{35}$$

The covariance kernel in these coordinates is $\log |v - \varepsilon u|^{-1}$. Since ν has mass zero, subtract $\log |v|^{-1}$. Multiply by fixed smooth cutoffs in u, v, equal to one on the supports, with the v cutoff bounded away from zero. The resulting two-variable kernel G_ε is smooth and compactly supported, and for each integer M its C^M norm is at most $C_M \varepsilon$. Indeed the two logarithms differ by $O(\varepsilon)$, uniformly with every fixed collection of derivatives, when $|v - \varepsilon u|$ is bounded below.

Periodize G_ε on a fixed sufficiently large cube. Its Fourier coefficients satisfy

$$|a_{m,n}| \le C_M \varepsilon (1 + |m| + |n|)^{-M}.$$

For a further fixed cutoff χ equal to one on the supports, Fourier duality and Cauchy–Schwarz give

$$|\nu(\chi e^{im\cdot u})| \le C\|\nu\|_- \|\chi e^{im\cdot u}\|_{\dot H^{d/2}} \le C' \|\nu\|_- (1 + |m|)^{d/2}.$$

The analogous estimate holds for μ. Choosing $M > 3d$ makes the double Fourier series absolutely summable after these two polynomial factors. Pairing this series with $\nu \otimes \mu$ proves (35). Approximation extends the estimate to the closed Gaussian Hilbert spaces generated by the annular fields.

For neighboring logarithmic scales use Cauchy–Schwarz instead. Only finitely many index differences are too small for (35); for the others the bound decays as $Cq^{|i-j|}$. Consequently there is $K \ge 1$, independent of the number of scales and of the numbers of coordinates, such that every finite Gaussian vector assembled from these blocks has covariance matrix satisfying

$$T \le K \operatorname{diag}(T_{11}, \dots, T_{nn}).$$

To check this matrix inequality, apply the scalar covariance bounds to arbitrary linear combinations within each block; the maximum absolute row sum of the resulting scalar bounds is uniformly finite.

Gaussian Hölder inequality [7, Theorem 1(i)] now gives

$$\mathbb{E} \prod_i F_i(X_i) \le \prod_i \left(\mathbb{E} F_i(X_i)^K \right)^{1/K}$$

for nonnegative measurable functions of finite coordinate vectors from the respective blocks. For functions of the entire restricted fields, apply this to conditional expectations with respect to increasing finite coordinate sets and use bounded martingale convergence. In particular, if $\mathbb{P}[E_i^c] \le \delta$, then for any subset I,

$$\mathbb{P}\left[\bigcap_{i \in I} E_i^c \right] \le \delta^{|I|/K}.$$

A union bound over subsets of failing indices yields

$$\mathbb{P}\left[\sum_{i=1}^{n} \mathbf{1}_{E_i} < \theta n\right] \leq 2^n \delta^{(1-\theta)n/K}.$$

Taking $\delta \leq \exp\{-K(A + \log 2)/(1-\theta)\}$ proves the assertion with $p = 1 - \delta$. □

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